

From Unstructured Collaboration to Synergistic Solidarity: Scaffolding Teamwork Skills, Values, and Virtues in Computing

Stephanie J. Lunn*
Florida International University
Miami, Florida, USA
sjlunn@fiu.edu

Maíra Marques Samary*
Boston College
Chestnut Hill, Massachusetts, USA
marquemo@bc.edu

Stephen Frezza*
Franciscan University of Steubenville
Steubenville, Ohio, USA
sfrezza@franciscan.edu

Priscilla Jimenez-Pazmino
University of St. Thomas
Saint Paul, Minnesota, USA
priscilla.jimenez@stthomas.edu

Janice L. Pearce
Berea College
Berea, Kentucky, USA
janpearce@berea.edu

Daniel Prol
University of Houston
Houston, Texas, USA
dprol@uh.edu

Shanon Reckinger
University of Illinois Chicago
Chicago, Illinois, USA
shanon@uic.edu

Michael James Scott
Falmouth University
Penryn, Cornwall, UK
michael.scott@falmouth.ac.uk

Carolin Wortmann
University of Münster
Münster, Germany
carolin.wortmann@uni-muenster.de

Batya Zamansky
University of Illinois
Urbana-Champaign
Urbana, Illinois, USA
batyaz2@illinois.edu

CCS Concepts

• Social and professional topics → Computing education.

Keywords

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*Working Group Co-Leader

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Table 1: Primary studies included in this systematic literature review (555 studies, 2016–2026)

Title	Authors	Year	Venue
A method to analyze computer science students' teamwork in online collaborative learning environments	Rebecca Vivian, Katrina Falkner, Nickolas Falkner et al.	2016	ACM Transactions on Computing Education
Academy-industry collaboration and the effects of the involvement of undergraduate students in real world activities	Elaine Venson, Rejane Figueiredo, Wander Silva et al.	2016	2016 IEEE Frontiers in Education Conference (FIE)
Applying scrum project management in ECE curriculum	Robert B. Bass, Branimir Pejcinovic, John Grant	2016	2016 IEEE Frontiers in Education Conference (FIE)
CapStone: a cloud-based platform for multi-party collaboration on capstone projects	Xiaocong Fan	2016	2016 IEEE Frontiers in Education Conference (FIE)
Collaboration and human factors in software development: teaching agile methodologies based on industrial insight	Martin Kropp, Andreas Meier	2016	2016 IEEE Global Engineering Education Conference (EDUCON)
Collaboration support in an international computer science capstone course	Robert Adams, Carsten Kleiner	2016	Social computing and social media
Developing diversity awareness of software engineers: a diversity framework and its application in an academic and life-long learning context	Christina Bohm, Renate Motschnig	2016	2016 IEEE Frontiers in Education Conference (FIE)
Distributed software development with one hand tied behind the back: a course unit to experience the role of communication in GSD	Marco Kuhrmann, Jurgen Munch	2016	2016 IEEE 11th International Conference on Global Software Engineering Workshops (ICGSEW)
Enhancing the students project with team based learning approach: a case study	Shivalingappa Battur, Mahesh S. Patil, Padmashree Desai et al.	2016	2016 IEEE 4th International Conference on MOOCs, Innovation and Technology in Education (MITE)
Evaluating student teams: do educators know what students think?	Anya Tafliovich, Andrew Petersen, Jennifer Campbell	2016	Proceedings of the 47th ACM Technical Symposium on Computing Science Education
Evaluation of teamwork competence acquisition by using CTMTC methodology and learning analytics techniques	Miguel Á. Conde, Francisco J. Rodríguez-Sedano, Lidia Sánchez-González et al.	2016	Proceedings of the Fourth International Conference on Technological Ecosystems for Enhancing Multiculturality
Evaluation of the CTMTC methodology for assessment of teamwork competence development and acquisition in higher education	Miguel Á. Conde, Ángel Hernández-García, Francisco J. García-Peñalvo et al.	2016	Learning and collaboration technologies
Experience report of teaching agile collaboration and values: agile software development in large student teams	Andreas Meier, Martin Kropp, Gerald Perellano	2016	2016 IEEE 29th International Conference on Software Engineering Education and Training (CSEET)
Experience report: guiding faculty & students to participate in humanitarian FOSS communities	Clifton Kussmaul	2016	2016 IEEE Eighth International Conference on Technology for Education (T4E)
Exploration of participation in student software engineering teams	Linda Marshall, Vreda Pieterse, Lisa Thompson et al.	2016	ACM Transactions on Computing Education
Green-lighting proposals for software engineering team-based project courses	Todd Sedano, Arthi Rengasamy, Cecile Peraire	2016	2016 IEEE 29th International Conference on Software Engineering Education and Training (CSEET)

Continued on next page

Table 1 – continued from previous page

Title	Authors	Year	Venue
Improving learning outcomes through systematic group reformation: the role of skills and personality in software engineering education	Amir Mujkanovic, Andreas Bollin	2016	Proceedings of the 9th International Workshop on Cooperative and Human Aspects of Software Engineering
Integrated development of technical and base competencies: fostering reflection skills in software engineers to be	Veronika Thurner, Kathrin Schlierkamp, Axel Bottcher et al.	2016	2016 IEEE Global Engineering Education Conference (EDUCON)
Investigation of a team-based activity support system used in a project-based learning course	Tessai Hayama, Saki Hayashi, Yuichi Kondo	2016	2016 11th International Conference on Knowledge, Information and Creativity Support Systems (KICSS)
Learning effectiveness of team discussions in various software engineering education courses	Masashi Shuto, Hironori Washizaki, Katsuhiko Kakehi et al.	2016	2016 IEEE 29th International Conference on Software Engineering Education and Training (CSEET)
Learning to program: gender differences and interactive effects of students' motivation, goals, and self-efficacy on performance	Alex Lishinski, Aman Yadav, Jon Good et al.	2016	Proceedings of the 2016 ACM Conference on International Computing Education Research
Project-based learning with examples from industry in university courses: an experience report from an undergraduate requirements engineering course	Marian Daun, Andrea Salmon, Thorsten Weyer et al.	2016	2016 IEEE 29th International Conference on Software Engineering Education and Training (CSEET)
Reflection is hard: teaching and learning reflective practice in a software studio	Julia Prior, Samuel Ferguson, John Leaney	2016	Proceedings of the Australasian Computer Science Week Multi-conference
Software project management as team building intervention	Mohammad Khalid Shaikh, Akhtar Raza, Kamran Ahsan	2016	Journal of Basic & Applied Sciences
Using the random forest classifier to assess and predict student learning of software engineering teamwork	Dragutin Petkovic, Marc Sosnick-Perez, Kazunori Okada et al.	2016	2016 IEEE Frontiers in Education Conference (FIE)
What communication tools students use in software projects and how do different tools suit different parts of project work?	Otto Seppälä, Tapio Auvinen, Ville Karavirta et al.	2016	Proceedings of the 38th International Conference on Software Engineering Companion
When teams go crazy: an environment to experience group dynamics in software project management courses	Marco Kuhrmann, Jürgen Münch	2016	Proceedings of the 38th International Conference on Software Engineering Companion
Why computing students learn on their own: motivation for self-directed learning of computing	Robert McCartney, Jonas Boustedt, Anna Eckerdal et al.	2016	ACM Transactions on Computing Education
A holistic capstone experience: beyond technical ability	Andrew Scott, William Krehling, Mark Holliday et al.	2017	Proceedings of the 18th Annual Conference on Information Technology Education
Agile learning: sprinting through the semester	Guido Lang	2017	Information Systems Education Journal
Agile manifesto for teaching and learning	Timothy C Krehbiel, Peter A Salzarulo, Michelle L Cosmah et al.	2017	Journal of Effective Teaching
Analysing and measuring students' experiences	Evangelia Krassadaki, Nikolaos F. Matsis	2017	2017 IEEE Global Engineering Education Conference (EDUCON)

Continued on next page

Table 1 – continued from previous page

Title	Authors	Year	Venue
Bad blood: managing toxic relationships through Belbin roles for first year software engineering students	Wai Sze Leung	2017	Proceedings of the 3rd International Conference on Communication and Information Processing
Blending communities and team-based learning in a programming course	Ignacio Cabrera, Jorge Villalon, Jorge Chavez	2017	IEEE Transactions on Education
CCNY joint senior design program in assistive technology across department boundaries	Zhigang Zhu, Jizhong Xiao	2017	2017 IEEE Integrated STEM Education Conference (ISEC)
Conflict problem solving in digital technology: a case study in virtual teamwork for online learning	Murty Magda Pane, Christian Siregar, Hari Sriyanto	2017	Proceedings of the 1st International Conference on Digital Technology in Education
Creation of educational games — project based learning in e-learning systems course	Mile Jovanov, Bojan Ilijoski, Emil Stankov et al.	2017	2017 IEEE Global Engineering Education Conference (EDUCON)
Cross-course project-based learning in requirements engineering: an eight-year retrospective	Joshua C. Nwokeji, Psem Stephen T. Frezza	2017	2017 IEEE Frontiers in Education Conference (FIE)
Cultural factors influencing international collaborative software engineering education in China	Yuqing Wang, Jouni Markkula, Jing Jiang	2017	2017 24th Asia-Pacific Software Engineering Conference (APSEC)
Deploying team-based learning at undergraduate software engineering courses	Santiago Matalonga, Gaston Mousques, Alejandro Bia	2017	2017 IEEE/ACM 1st International Workshop on Software Engineering Curricula for Millennials (SECM)
Don't underestimate the human factors! Exploring team communication effects	Fabian Kortum, Jil Klünder, Kurt Schneider	2017	Product-focused software process improvement
Enhancing collaboration among undergraduates in informatics: a teaching and learning process based on crowdsourcing	William Simao De Deus, Heydi Miura Machado, Renata Marques Barros et al.	2017	2017 IEEE Frontiers in Education Conference (FIE)
Enhancing engineering education with wikispaces	Udai Kumar Kudikyala	2017	2017 International Conference on Transforming Engineering Education (ICTEE)
Evolving a project-based software engineering course: a case study	David Delgado, Alejandro Velasco, Jairo Aponte et al.	2017	2017 IEEE 30th Conference on Software Engineering Education and Training (CSEE&T)
Experimenting with realism in software engineering team projects: an experience report	Robbie Simpson, Tim Storer	2017	2017 IEEE 30th Conference on Software Engineering Education and Training (CSEE&T)
From diversity by numbers to diversity as process: supporting inclusiveness in software development teams with brainstorming	Anna Filippova, Erik Trainer, James D. Herbsleb	2017	2017 IEEE/ACM 39th International Conference on Software Engineering (ICSE)
Hardening freshman engineering student soft skills	Andrea Borowczak, Mike Borowczak	2017	2017 FYEE Conference Proceedings
Improving collaborative learning in online software engineering education	Colin J. Neill, Joanna F. DeFranco, Raghvinder S. Sangwan	2017	European Journal of Engineering Education
Improving communication in multicultural teams — a web-based model and its application in project management education	Renate Motschnig, Seyhan Guver	2017	2017 IEEE Frontiers in Education Conference (FIE)
Integrated E-project collaborative management system: empirical study for problem-based learning project	Kemachart Kemavuthanon	2017	2017 9th International Conference on Information Technology and Electrical Engineering (ICITEE)

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Table 1 – continued from previous page

Title	Authors	Year	Venue
Interdisciplinary software projects as an active methodology to practice for the profession	Maria Augusta Vieira Nelson, Rommel Vieira Carneiro, Marco Rodrigo Costa	2017	2017 IEEE/ACM 1st International Workshop on Software Engineering Curricula for Millennials (SECM)
Investigating students' metacognitive skills while working on information systems development projects	Harry Budi Santoso, Oenardi Lawanto, Betty Purwandari et al.	2017	2017 7th World Engineering Education Forum (WEEF)
Investigating undergraduate students' communication self-efficacy during an engineering design course	Deeksha Seth, Gabriel N. Carryon	2017	2017 IEEE Frontiers in Education Conference (FIE)
Lean learning - applying lean techniques to improve software engineering education	Robert Chatley, Tony Field	2017	2017 IEEE/ACM 39th International Conference on Software Engineering: Software Engineering Education and Training Track (ICSE-SEET)
Learning agile with tech startup software engineering projects	Kevin Buffardi, Colleen Robb, David Rahn	2017	Proceedings of the 2017 ACM Conference on Innovation and Technology in Computer Science Education
Learning scrum by doing real-life projects	Monica Villavicencio, Erika Narvaez, Edgar Izquierdo et al.	2017	2017 IEEE Global Engineering Education Conference (EDUCON)
Lifting the constraints — closing the skills gap with authentic student projects	Richa Awasthy, Shayne Flint, Ramesh Sankaranarayana	2017	2017 IEEE Global Engineering Education Conference (EDUCON)
Meta-assessment in a project-based systems engineering course	Niva Wengrowicz, Yehudit Judy Dori, Dov Dori	2017	Assessment & Evaluation in Higher Education
On a pursuit for perfecting an undergraduate requirements engineering course	Chandan R. Rupakheti, Mark Hays, Sriram Mohan et al.	2017	2017 IEEE 30th Conference on Software Engineering Education and Training (CSEE&T)
Project-based learning in an international virtual classroom to teach global software engineering	Jesus Manuel Olivares-Ceja, Brandon Guierrez Sanchez, Patricia Brockmann et al.	2017	EDULEARN17 Proceedings
Relationship between the five factor model personality and learning effectiveness of teams in three information systems education courses	Masashi Shuto, Hironori Washizaki, Katsuhiko Kakehi et al.	2017	2017 18th IEEE/ACIS International Conference on Software Engineering, Artificial Intelligence, Networking and Parallel/Distributed Computing (SNPD)
Student perspectives of team-based learning in a CS course: summary of qualitative findings	Michael S. Kirkpatrick	2017	Proceedings of the 2017 ACM SIGCSE Technical Symposium on Computer Science Education
Study of motivation of engineering students on multinational design projects	Jorge Rodriguez, Ivan Esparragoza, Jared R. Ocampo et al.	2017	IEEE Revista Iberoamericana de Tecnologías del Aprendizaje
Teaching software engineering course with cooperative learning method: a pilot study	Shuib Basri, Aliza Sarlan, Norshakirah A. Aziz et al.	2017	2017 7th World Engineering Education Forum (WEEF)
Teaching software project management using project based learning (PBL) and group projects	Jeremy Straub, Scott Kerlin, David Whalen	2017	2017 IEEE International Conference on Electro Information Technology (EIT)

Continued on next page

Table 1 – continued from previous page

Title	Authors	Year	Venue
The archetype learning method — scaffolding teamwork competences in the engineering classroom	Arnold Pears, Mats Daniels, Asa Cajander	2017	2017 IEEE Frontiers in Education Conference (FIE)
Towards an approach to prevent social loafing in software development teams	Ilenia Fronza, Xiaofeng Wang	2017	2017 ACM/IEEE International Symposium on Empirical Software Engineering and Measurement (ESEM)
Use of gamification to teach agile values and collaboration: a multi-week scrum simulation project in an undergraduate software engineering course	Sonja Hof, Martin Kropp, Marla Landolt	2017	Proceedings of the 2017 ACM Conference on Innovation and Technology in Computer Science Education
Use of role-play and gamification in a software project course	Bruce R. Maxim, Stein Brunvand, Adrienne Decker	2017	2017 IEEE Frontiers in Education Conference (FIE)
What can students get from a software engineering capstone course?	Maria Cecilia Bastarrica, Daniel Perovich, Maira Marques Samary	2017	2017 IEEE/ACM 39th International Conference on Software Engineering: Software Engineering Education and Training Track (ICSE-SEET)
You want me to work with Who?: stakeholder perceptions of automated team formation in project-based courses	Farnaz Jahanbakhsh, Wai-Tat Fu, Karrie Karahalios et al.	2017	Proceedings of the 2017 CHI Conference on Human Factors in Computing Systems
A case study of using scrum in teaching software process	Minjie Hu, Sandra Cleland, Aaron Steele	2018	Proceedings of the 9th Annual Conference of Computing and Information Technology Research and Education New Zealand (CITRENZ 2018)
A hackathon methodology for undergraduate course projects	Kiev Gama, Breno Alencar, Filipe Calegario et al.	2018	2018 IEEE Frontiers in Education Conference (FIE)
A scalable methodology to guide student teams executing computing projects	Jeffrey S. Saltz, Robert R. Heckman	2018	ACM Transactions on Computing Education
A study of team formation strategies and their impact on individual student learning using educational data mining (EDM)	Vijayalakshmi M, Suman Salimath, Ashok S. Shettar et al.	2018	2018 IEEE Tenth International Conference on Technology for Education (T4E)
Acquiring professional software engineering skills through studio-based learning	Daniela Rosca	2018	2018 17th International Conference on Information Technology Based Higher Education and Training (ITHET)
Active project based learning pedagogies: learning hardware, software design and wireless sensor instrumentation	Fahd Ahmed Banakhr, Muhammad Javed Iqbal, Nabil Shaukat	2018	2018 IEEE Global Engineering Education Conference (EDUCON)
Agile practices and the promotion of entrepreneurial skills in software development	Cristiano Tolfo, Raul Sidnei Wazlawick, Marcelo Gitirana Gomes Ferreira et al.	2018	Journal of Software: Evolution and Process
An activity-based undergraduate software engineering course to engage students and encourage learning	Eileen T. Kraemer, Murali Sitaraman, Joseph Hollingsworth	2018	Proceedings of the 3rd European Conference of Software Engineering Education
An undergraduate project combining computer science and the arts: an experience report of a multidisciplinary capstone design	Sixto J. Castro, Luis Jácome, Ana Gabriela Concha et al.	2018	Proceedings of the 7th Computer Science Education Research Conference

Continued on next page

Table 1 – continued from previous page

Title	Authors	Year	Venue
Analyzing main characteristics in software engineering projects	Jakeline Marcos-Abed	2018	2018 IEEE Frontiers in Education Conference (FIE)
approach for the determination and correlation of diversity and efficiency of software development teams	Seher Altiner, Mustafa B. Ayhan	2018	South African Journal of Science
Assessing programming education approaches for college beginners with different backgrounds	Hong Liu, Chen Zhong	2018	Proceedings of the 10th International Conference on Education Technology and Computers
Beyond surveys: analyzing software development artifacts to assess teaching efforts	Christoph Matthies, Ralf Teusner, Guenter Hesse	2018	2018 IEEE Frontiers in Education Conference (FIE)
Bringing together undergraduate and postgraduate students in software engineering team project: experiences and lessons	Georgia M. Kapitsaki, Styliani Kleanthous Loizou	2018	Proceedings of the 23rd Annual ACM Conference on Innovation and Technology in Computer Science Education
Collaborative agile learning in online environments: strategies for improving team regulation and project management	Ingrid Noguera, Ana-Elena Guerrero-Roldán, Ricard Masó	2018	Computers & Education
Collaborative and teamwork software development in an undergraduate software engineering course	Claudia Raibulet, Francesca Arcelli Fontana	2018	Journal of Systems and Software
Combining agile software development and service-learning: a case study in experiential IS education	Spencer Robinson, Margeret Hall	2018	Proceedings of the 49th ACM Technical Symposium on Computer Science Education
Continuous delivery of personalized assessment and feedback in agile software engineering projects	Xiaoying Bai, Mingjie Li, Dan Pei et al.	2018	ICSE '18: 40th International Conference on Software Engineering
Continuous delivery of personalized assessment and feedback in agile software engineering projects	Xiaoying Bai, Mingjie Li, Dan Pei et al.	2018	Proceedings of the 40th International Conference on Software Engineering: Software Engineering Education and Training
Effective approach in making capstone project a holistic learning experience to students of undergraduate computer science engineering program	Deepamala N, Shobha G	2018	Journal of Technology and Science Education
Enhancing software engineering education through experimentation: an experience report	Marco Kuhrmann, Jurgen Munch	2018	2018 IEEE International Conference on Engineering, Technology and Innovation (ICE/ITMC)
Enhancing the student learning experience in software engineering project courses	Maira Marques, Sergio F. Ochoa, Maria Cecilia Bastarrica et al.	2018	IEEE Transactions on Education
Equal but not equitable: self-reported data obscures gendered differences in project teams	Laura J. Hirshfield	2018	IEEE Transactions on Education
Evaluating the influence of PBL on the development of soft skills in a computer engineering undergraduate program	Felipe L. F. Barros, Roberto A. Bittencourt	2018	2018 IEEE Frontiers in Education Conference (FIE)
Exploring collaborative writing of user stories with multimodal learning analytics: a case study on a software engineering course	Rene Noel, Fabian Riquelme, Roberto Mac Lean et al.	2018	IEEE Access

Continued on next page

Table 1 – continued from previous page

Title	Authors	Year	Venue
Exploring students' perceptions of service-learning experiences in an undergraduate web design course	Sang Joon Lee, Charlie Wilder, Chien Yu	2018	Teaching in Higher Education
Fostering 21 century learning and innovation competencies through students' online collaborative activities in software engineering courses	Alejandro Adorjan, Miguel Nunez-Del-Prado	2018	2018 IEEE World Engineering Education Conference (EDUNINE)
Fostering soft skills in project-oriented learning within an agile atmosphere	Hadas Chassidim, Dani Almog, Shlomo Mark	2018	European Journal of Engineering Education
Fostering teamwork with an online idea stock exchange	Lori Scarlatos, Tony Scarlatos	2018	Journal of Computing Sciences in Colleges
From explaining how random forest classifier predicts learning of software engineering teamwork to guidance for educators	Dragutin Petkovic, Sabiha H Barlaskar, Jizhou Yang et al.	2018	2018 IEEE Frontiers in Education Conference (FIE)
Gendered influences on peer assessments of team behavior	Emily Miller, Reid Bailey	2018	2018 IEEE Frontiers in Education Conference (FIE)
Gendered risks of team-based learning: a model of inequitable task allocation in project-based learning	Robin R. Fowler, Magel P. Su	2018	IEEE Transactions on Education
How does participating in a capstone project with industrial customers affect student attitudes?	Maria Paasivaara, Dragoş Vodă, Ville T. Heikkilä et al.	2018	Proceedings of the 40th International Conference on Software Engineering: Software Engineering Education and Training
How personality diversity influences team performance in student software engineering teams	Vreda Pieterse, Mpho Leeu, Marko Van Eekelen	2018	2018 Conference on Information Communications Technology and Society (ICTAS)
Impact of experiential learning on students' success in undergraduate engineering	Padmashree Desai, Achyut Bhandiwad, Ashok S. Shettar	2018	2018 IEEE 18th International Conference on Advanced Learning Technologies (ICALT)
Implementation of a PBL/CDIO methodology at ISEP-P.PORTO systems engineering course	Jose Manuel Magalhaes, Alberto Peixoto Pinto, Maria Teresa Costa et al.	2018	2018 3rd International Conference of the Portuguese Society for Engineering Education (CISPÉE)
Improving electronics engineering students' skills by projects' college competition	Moath Awawdeh, Tarig Faisal, Fatima Fadel et al.	2018	2018 Advances in Science and Engineering Technology International Conferences (ASET)
Improving retention and confidence through cross-course collaborative project-based learning	Jennifer Winikus, Lukasz Ziarek, Carl Alphonse et al.	2018	2018 IEEE Frontiers in Education Conference (FIE)
Inquiry-based electronics laboratory: students' attitudes	Aharon Gero, Beto Catz, Nissim Sabag	2018	2018 28th EAEEIE Annual Conference (EAEEIE)
Integrating project based learning and project management for software engineering teaching: an experience report	Maria Lydia Fioravanti, Bruno Sena, Leo Natan Paschoal et al.	2018	Proceedings of the 49th ACM Technical Symposium on Computer Science Education
Investigating the effects of personality on software design in a higher education setting through an experiment	Hector G. Perez-Gonzalez, Alberto Nunez-Varela, Francisco E. Martinez-Perez et al.	2018	2018 6th International Conference in Software Engineering Research and Innovation (CONISOFT)
Involving external stakeholders in project courses	Jan-Philipp Steghöfer, Håkan Burden, Regina Hebig et al.	2018	ACM Transactions on Computing Education

Continued on next page

Table 1 – continued from previous page

Title	Authors	Year	Venue
Long term effects of pair programming	Max O. Smith, Andrew Giugliano, Andrew DeOrio	2018	IEEE Transactions on Education
More than a feeling: applying a data-driven framework in the technical and professional communication team project	Chris Lam	2018	IEEE Transactions on Professional Communication
Motivation and attitude of computer engineering students toward soft skills	Mariecke Schipper, Esther Van Der Stap-pen	2018	2018 IEEE Global Engineering Education Conference (EDUCON)
Moving from waterfall to agile process in software engineering capstone projects	Mohammad Alshayeb, Sajjad Mahmood, Khalid Aljasser	2018	Computer Science & Information Technology
Pilot experience applying an active learning methodology in a software engineering classroom	Alicia Garcia-Holgado, Francisco J. Garcia-Penalvo, Maria Jose Rodriguez-Conde	2018	2018 IEEE Global Engineering Education Conference (EDUCON)
Problem solving to teach advanced algorithms in heterogeneous groups	Florent Bouchez-Tichadou	2018	Proceedings of the 23rd Annual ACM Conference on Innovation and Technology in Computer Science Education
Problem-based learning. An experience on the inclusion of quality problems in educational software engineering	Yasnalla Rivero Pena, Jesus Rafael Hechavarria Hernandez, Maikel Leyva Vazquez	2018	2018 World Engineering Education Forum - Global Engineering Deans Council (WEEF-GEDC)
Qualitative findings from an online course on machine learning	Steve Chenoweth, Panagiotis K. Linos	2018	2018 IEEE Frontiers in Education Conference (FIE)
Redesigning an undergraduate software engineering course for a large cohort	Claudia Iacob, Shamal Faily	2018	Proceedings of the 40th International Conference on Software Engineering: Software Engineering Education and Training
Reflections on 17 years of ICT capstone project coordination: effective strategies for managing clients, teams and assessment	Nicole Herbert	2018	Proceedings of the 49th ACM Technical Symposium on Computer Science Education
Scrum2kanban: integrating kanban and scrum in a university software engineering capstone course	Christoph Matthies	2018	Proceedings of the 2nd International Workshop on Software Engineering Education for Millennials
Seven Principles of Undergraduate Capstone Project Management	Xiaocong Fan	2018	Proceedings of the International Conference on Software Engineering Research and Practice (SERP'18)
SimulES-W: a collaborative game to improve software engineering teaching	Elizabeth Suescún, Julio Cesar Camapio Do Prado Leite, Vera Werneck et al.	2018	Computación y Sistemas
Skill-based group allocation of students for project-based learning courses using genetic algorithm: weighted penalty model	Ravneil Nand, Anuraganand Sharma, Karuna Reddy	2018	2018 IEEE International Conference on Teaching, Assessment, and Learning for Engineering (TALE)
Software engineering for millennials, by millennials	Jocelyn Simmonds, Máira Marques Samary, Milenko Tomic et al.	2018	Proceedings of the 2nd International Workshop on Software Engineering Education for Millennials

Continued on next page

Table 1 – continued from previous page

Title	Authors	Year	Venue
Software engineering problems and their relationship to perceived learning and customer satisfaction on a software capstone project	Jari Vanhanen, Timo O.A. Lehtinen, Casper Lassenius	2018	Journal of Systems and Software
Structure or nurture?: the effects of team-building activities and team composition on team outcomes	Emily M. Hastings, Farnaz Jahanbakhsh, Karrie Karahalios et al.	2018	Proceedings of the ACM on Human-Computer Interaction
Student learning in computing-based service-learning	Nusaybah Abu-Mulaweh, William Oakes	2018	2018 IEEE Frontiers in Education Conference (FIE)
TEAMSCOPE: measuring software engineering processes with teamwork telemetry	An Ju, Armando Fox	2018	Proceedings of the 23rd Annual ACM Conference on Innovation and Technology in Computer Science Education
Tech startup learning activities: a formative evaluation	Kevin Buffardi	2018	Proceedings of the 2nd International Workshop on Software Engineering Education for Millennials
The adoption of open source projects in engineering education: a real software development experience	Debora M. C. Nascimento, Christina F. G. Chavez, Roberto A. Bittencourt	2018	2018 IEEE Frontiers in Education Conference (FIE)
The hard work of soft skills: augmenting the project-based learning experience with interdisciplinary teamwork	Jane S. Vogler, Penny Thompson, David W. Davis et al.	2018	Instructional Science
Toward increasing collaboration awareness in software engineering teams	Shuddha Chowdhury, Charles Walter, Rose Gamble	2018	2018 IEEE Frontiers in Education Conference (FIE)
Towards team formation using belbin role types and a social networks analysis approach	Josue-Miguel Flores-Parra, Manuel Castafon-Puga, Richard David Evans et al.	2018	2018 IEEE Technology and Engineering Management Conference (TEMSCON)
‘Helpful,’ ‘irritating,’ and ‘smart’: student perspectives on teams in a mechanical engineering program	Nancy B. Barr, James P. De Clerck	2018	2018 IEEE International Professional Communication Conference (ProComm)
A comparison of peer rating behavior between students in hard vs. Soft disciplines	Maizey Benner, Daniel Ferguson, Behzad Beigpourian et al.	2019	2019 IEEE Frontiers in Education Conference (FIE)
A method to provide student peer mentorship within the capstone experience	David E. Schmidt	2019	2019 IEEE Frontiers in Education Conference (FIE)
Actions to promote diversity in engineering studies: a case study in a computer science degree	Alicia Garcia-Holgado, Andrea Vazquez-Ingelmo, Sonia Verdugo-Castro et al.	2019	2019 IEEE Global Engineering Education Conference (EDUCON)
Assessing students’ IT professional values in a global project setting	S. Frezza, M. Daniels, A. Wilkin	2019	ACM Transactions on Computing Education
Balancing theory and practice in software engineering education – a PBL, toolset based approach	Awdren Fontao, Bruno Gadelha, Alberto Castro Junior	2019	2019 IEEE Frontiers in Education Conference (FIE)
Behavior-driven dynamics in agile development: the effect of fast feedback on teams	Fabian Kortum, Jil Klunder, Kurt Schneider	2019	2019 IEEE/ACM International Conference on Software and System Processes (ICSSP)
Boosting student performance with peer reviews; integration and analysis of peer reviews in a gamified software engineering classroom	Kay Berkling, Katja Neubehler	2019	2019 IEEE Global Engineering Education Conference (EDUCON)

Continued on next page

Table 1 – continued from previous page

Title	Authors	Year	Venue
Case study: using project based learning to develop parallel programming and soft skills	Awad A Younis, Rajshekhar Sunderraman, Mike Metzler et al.	2019	2019 IEEE International Parallel and Distributed Processing Symposium Workshops (IPDPSW)
Characterizing capstone design teaching: a functional taxonomy	James J. Pembridge, Marie C. Parette	2019	Journal of Engineering Education
Collaborating with industrial customers in a capstone project course: the customers' perspective	Maria Paasivaara, Jari Vanhanen, Casper Lassenius	2019	2019 IEEE/ACM 41st International Conference on Software Engineering: Software Engineering Education and Training (ICSE-SEET)
Continuous assessment in software engineering project course using publicly available data from GitHub	Henrik Gustavsson, Marcus Brohede	2019	Proceedings of the 15th International Symposium on Open Collaboration
Design of a course oriented to the comprehension of agile methods based on teamwork, role-play, and class project with a real client	Alix E. Rojas, Camilo Mejia-Moncayo	2019	2019 International Conference on Virtual Reality and Visualization (ICVRV)
Design of capstone projects under CDIO mode	Wei Zhang	2019	2019 International Conference on Computational Science and Computational Intelligence (CSCI)
Designing a multi-disciplinary group project for computer science and engineering students	Kheng Leong Tan, Woon Boon Goh	2019	2019 IEEE Global Engineering Education Conference (EDUCON)
Developing soft and technical skills through multi-semester, remotely mentored, community-service projects	Janet Davis, Samuel A. Rebelsky	2019	Proceedings of the 50th ACM Technical Symposium on Computer Science Education
Does social sensitivity impact virtual teams?	Lisa L. Lacher, Cydnee Biehl	2019	Proceedings of the 50th ACM Technical Symposium on Computer Science Education
Effect of controlling group heterogeneity on student performance in a graphical programming course	Evelia Y. Salinas, Adrienne E. Williams, Christine E. King	2019	2019 IEEE Frontiers in Education Conference (FIE)
Exploring the gap between the student expectations and the reality of teamwork in undergraduate software engineering group projects	Claudia Iacob, Shamal Faily	2019	Journal of Systems and Software
Facilitating entrepreneurial experiences through a software engineering project course	Håkan Burden, Jan-Philipp Steghöfer, Oskar Hagvall Svensson	2019	2019 IEEE/ACM 41st International Conference on Software Engineering: Software Engineering Education and Training (ICSE-SEET)
Gamification in software engineering education	G. Ivanova, V. Kozov, P. Zlatarov	2019	2019 42nd International Convention on Information and Communication Technology, Electronics and Microelectronics (MIPRO)
How much authenticity can be achieved in software engineering project based courses?	Zahra Shakeri Hossein Abad, Muneera Bano, Didar Zowghi	2019	2019 IEEE/ACM 41st International Conference on Software Engineering: Software Engineering Education and Training (ICSE-SEET)
Impact of using agile methods in software engineering education: a case study	Serein Al-Ratrout	2019	2019 6th International Conference on Control, Decision and Information Technologies (CoDIT)

Continued on next page

Table 1 – continued from previous page

Title	Authors	Year	Venue
Industry-oriented project-based learning of software engineering	Maria Spichkova	2019	2019 24th International Conference on Engineering of Complex Computer Systems (ICECCS)
Industry-university collaboration: an educational program with automotive industry	Hector Morano-Okuno, Guillermo Sandoval-Benitez, Rafael Caltenco-Castillo et al.	2019	2019 IEEE International Conference on Engineering, Technology and Education (TALE)
Integrating introductory data science into computer and information literacy through collaborative project-based learning	Qiong Cheng, Felix Lopez, Athina Hadjixenofontos	2019	2019 IEEE Frontiers in Education Conference (FIE)
Investigating the role choice of female students in a software engineering team project	Laura Heels, Marie Devlin	2019	Proceedings of the 3rd Conference on Computing Education Practice
Managing diversity in distributed software development education—a longitudinal case study	Ivana Bosnić, Federico Ciccozzi, Ivica Crnković et al.	2019	ACM Transactions on Computing Education
Meta-heuristic approaches to tackle skill based group allocation of students in project based learning courses	Ravneil Nand, Anuraganand Sharma	2019	2019 IEEE Congress on Evolutionary Computation (CEC)
Mistakes in UML diagrams: analysis of student projects in a software engineering course	Stanislav Chren, Barbora Buhnova, Martin Macak et al.	2019	2019 IEEE/ACM 41st International Conference on Software Engineering: Software Engineering Education and Training (ICSE-SEET)
Pair programming as a didactical approach in higher education and its evaluation	Carolyn Gold-Veerkamp, Marco Klopp, Jorg Abke	2019	2019 IEEE Global Engineering Education Conference (EDUCON)
Participating in an industry based social service program: a report of student perception of what they learn and what they need	Miguel Ehécatl Morales Trujillo, Gabriel Alberto García Mireles	2019	2019 Federated Conference on Computer Science and Information Systems (FedCSIS)
Practical implementation of agile approaches in teaching process	Serein Al-Ratrout	2019	International Journal of Engineering and Advanced Technology
Practical method of improving the teamwork of engineering students using team contracts to minimize conflict situations	Maria Luisa Pertegal-Felices, Andres Fuster-Guillo, Maria Luisa Rico-Soliveres et al.	2019	IEEE Access
Problem-based learning: student perceptions of its value in developing professional skills for engineering practice	Úna Beagon, Dervilla Niall, Eabhannat Ní Fhloinn	2019	European Journal of Engineering Education
Project work division in agile distributed student teams - who develops what?	Ivana Bosnic, Igor Cavrak	2019	2019 ACM/IEEE 14th International Conference on Global Software Engineering (ICGSE)
Prototyping for internet of things with web technologies: a case on project-based learning using scrum	Bahtijar Vogel, Bo Peterson, Blerim Emruli	2019	2019 IEEE 43rd Annual Computer Software and Applications Conference (COMPSAC)
Relax, it's a game: utilizing gamification in learning agile scrum software development	Nitin Naik, Paul Jenkins	2019	2019 IEEE Conference on Games (CoG)
Research and practice of applying adaptive learning in computer science and IT degree programs	Yanzhen Qu, Richard Cai, Mazen Haj-Hussein	2019	2019 IEEE Frontiers in Education Conference (FIE)
Student assessment in a capstone computing course	Charles C Tappert, Avery M Leider, Sukun Li	2019	Proceedings of the 49th Annual Meeting of the Southeast Decision Sciences Institute

Continued on next page

Table 1 – continued from previous page

Title	Authors	Year	Venue
Student engagement in active learning software engineering courses	Bruce R. Maxim, Adrienne Decker, Jeffrey J. Yackley	2019	2019 IEEE Frontiers in Education Conference (FIE)
Students evaluation in AGILE after completing software engineering course	Sabbir M. Saleh, Supta Richard Philip	2019	2019 1st International Conference on Advances in Science, Engineering and Robotics Technology (ICASERT)
Students perception on the use of project-based learning in software engineering education	Mauricio Souza, Renata Moreira, Eduardo Figueiredo	2019	Proceedings of the XXXIII Brazilian Symposium on Software Engineering
Teaching an undergraduate software engineering course using active learning and open source projects	Mohsen Dorodchi, Erfan Al-Hossami, Mohammad Nagahisarchoghaei et al.	2019	2019 IEEE Frontiers in Education Conference (FIE)
Teaching generic competences in software engineering via E-learning	Marco Klopp	2019	2019 IEEE International Conference on Engineering, Technology and Education (TALE)
Teaching software engineering tools to undergraduate students	Claudia Raibulet, Francesca Arcelli Fontana, Ilaria Pigazzini	2019	Proceedings of the 11th International Conference on Education Technology and Computers
Teaching software engineering with free open source software development: an experience report	Anya Tafilovich, Francisco Estrada, Thomas Caswell	2019	Proceedings of the 52nd Hawaii International Conference on System Sciences (HICSS 2019)
Teaching software maintenance	Keith Gallagher, Mark Fioravanti, Suzanne Kozaitis	2019	2019 IEEE International Conference on Software Maintenance and Evolution (ICSME)
Teamwork attitudes, interest and self-efficacy between online and face-to-face information technology students	Abdullah Konak, Sadan Kulturel-Konak, Gordon W. Cheung	2019	Team Performance Management: An International Journal
The effect of preferred leadership role and preferred team leadership structure on students' perception of team processes and outcome	Behzad Beigpourian, Hossein EbrahimiNejad, Mathew W. Ohland et al.	2019	2019 IEEE Frontiers in Education Conference (FIE)
The “bachelor project”: project based computer science education	Jan Renz, Christoph Meinel	2019	2019 IEEE Global Engineering Education Conference (EDUCON)
Towards a better understanding of team-driven dynamics in agile software projects: a characterization and visualization support in JIRA	Fabian Kortum, Oliver Karras, Jil Klünder et al.	2019	Product-focused software process improvement
Training software engineers using open-source software: the students' perspective	Gustavo Pinto, Clarice Ferreira, Cleice Souza et al.	2019	2019 IEEE/ACM 41st International Conference on Software Engineering: Software Engineering Education and Training (ICSE-SEET)
Using CATME to document and improve the effectiveness of teamwork in capstone courses	Behzad Beigpourian, Daniel M. Ferguson, Frederick C. Berry et al.	2019	2019 ASEE Annual Conference & Exposition
What did CS students recognize as study difficulties?	Ville Hamalainen, Ville Isomottonen	2019	2019 IEEE Frontiers in Education Conference (FIE)
Work in progress: self-directed approach for project based learning activity	Boon-Seng Chew, Chew-Lin Chia, Francis Teo	2019	2019 IEEE Global Engineering Education Conference (EDUCON)
A learning analytics tool for the analysis of students' telegram messages in the context of teamwork virtual activities	Miguel Á. Conde, Francisco J. Rodríguez-Sedano, Camino Fernández et al.	2020	Eighth International Conference on Technological Ecosystems for Enhancing Multiculturality

Continued on next page

Table 1 – continued from previous page

Title	Authors	Year	Venue
A methodology for psycho-biological assessment of stress in software engineering	Jan-Peter Ostberg, Daniel Graziotin, Stefan Wagner et al.	2020	PeerJ Computer Science
A model of creativity based learning for computer teaching to enhance creative skills of undergraduate students	Nopparat Meeplat	2020	Proceedings of the 2020 the 3rd International Conference on Computers in Management and Business
A project-based learning approach for enhancing learning skills and motivation in software engineering	Beatriz Pérez, Ángel L. Rubio	2020	Proceedings of the 51st ACM Technical Symposium on Computer Science Education
A transactive memory system perspective in software engineering education	Georgia M. Kapitsaki, Styliani Loizou Kleanthous, George A. Papadopoulos	2020	IEEE Transactions on Education
Active and collaborative learning based dynamic instructional approach in teaching introductory computer science course with python programming	Md Mahmudur Rahman, Monir H. Sharker, Roshan Paudel	2020	2020 IEEE Integrated STEM Education Conference (ISEC)
Assessing individual contributions to software engineering projects with git logs and user stories	Kevin Buffardi	2020	Proceedings of the 51st ACM Technical Symposium on Computer Science Education
Blended academic international mobility: tearing down barriers to mobility in a sustainable way	Nuno Escudeiro, Ana Barata, Paula Escud-eiro et al.	2020	2020 IEEE Global Engineering Education Conference (EDUCON)
Building psychological safety through training interventions: manage the team, not just the project	Lisa Dusenberry, Joy Robinson	2020	IEEE Transactions on Professional Communication
Collaborative learning in computing education: faculty perspectives and practices	Sharon Mason	2020	Proceedings of the 2020 ACM Conference on International Computing Education Research
Demonstrating the impact of international collaborative disciplinary experiences on student global, international, and intercultural competencies	M.S. Rob Elliott, Xiao Luo	2020	2020 IEEE Frontiers in Education Conference (FIE)
Differences in the students' achievements between traditional and project-based learning of basic engineering competencies: a quasi-experimental study	D. Purkovic, M. Prihoda	2020	2020 43rd International Convention on Information, Communication and Electronic Technology (MIPRO)
Different perspectives on students' presentation skills: teacher, peer, team and self-assessment	Sanja Candrljic	2020	2020 15th International Conference on Computer Science & Education (ICCSE)
Evaluating the benefits of team-based learning in a systems programming class	Alark Joshi, Marissa Schmidt, Shane Panter et al.	2020	2020 IEEE Frontiers in Education Conference (FIE)
Experience report on key success factors for promoting students' engagement in software development group projects	Ruth Schorr	2020	2020 IEEE World Conference on Engineering Education (EDU-NINE)
Experiments with multinational cross-course project	Mehul S Raval, Tolga Kaya, Mazad Zaveri et al.	2020	2020 IEEE International Conference on Teaching, Assessment, and Learning for Engineering (TALE)

Continued on next page

Table 1 – continued from previous page

Title	Authors	Year	Venue
Exploiting group shuffling dynamics to convey the importance of good software design	Jose I. Benedetto, Jaime Navon	2020	Proceedings of the ACM/IEEE 42nd International Conference on Software Engineering: Software Engineering Education and Training
Impact of team formation method on student performance, attitudes, and behaviors	Martiqua L Post, Anthony Barrett, Marlyse Williams et al.	2020	Journal of the Scholarship of Teaching and Learning
Improving multidisciplinary understanding through interdisciplinary project-based learning in a first-year orientation course	Megan McCormick, Jennifer Parham-Mocello, Donald Heer	2020	2020 IEEE Frontiers in Education Conference (FIE)
Investigating the impact and pedagogical implications of extroversion on team dynamics in post-secondary student teams	James A. Collier, Robin R. Fowler, Laura K. Alford	2020	2020 IEEE Frontiers in Education Conference (FIE)
Involving IT professionals in scrum student teams: an empirical study on the impact of students' learning	Panagiotis K. Linos, Ryan Rybarczyk, Nathan Partenheimer	2020	2020 IEEE Frontiers in Education Conference (FIE)
Measure students' contribution in web programming projects by exploring source code repository	Bao-An Nguyen, Kuan-Yu Ho, Hsi-Min Chen	2020	2020 International Computer Symposium (ICS)
Measuring students' contributions in software development projects using git metrics	Sivana Hamer, Christian Quesada-Lopez, Alexandra Martinez et al.	2020	2020 XLVI Latin American Computing Conference (CLEI)
Moving to project-based learning at the program level: an experience report	Isabel Sofia Brito, Joao Paulo Barros, Elsa Rodrigues	2020	2020 IEEE Global Engineering Education Conference (EDUCON)
Relation of individual time management practices and time management of teams	Tapio Auvinen, Nickolas Falkner, Arto Hellas et al.	2020	2020 IEEE Frontiers in Education Conference (FIE)
Scrum higher education – the scrum master supports as solution-focused coach	Martina Muller-Amthor, Georg Hagel, Matthias Gensheimer et al.	2020	2020 IEEE Global Engineering Education Conference (EDUCON)
Student feedback for globally distributed team application development	David Drohan, Patrick Seeling	2020	2020 IEEE Frontiers in Education Conference (FIE)
Success factors of IT projects: undergraduate students perspective	Muhammad Nazrul Islam, Anika Binte Islam	2020	2020 61st International Scientific Conference on Information Technology and Management Science of Riga Technical University (ITMS)
Teaching agile software engineering practices using scrum and a lowcode development platform-a case study	Jc Metrolho, Fr Ribeiro, P Passao	2020	Proceedings of the 15th Conference on Software Engineering Advances (pp. 160-165).
Teaching strategies in software engineering towards industry interview preparedness	William Gregory Johnson, Raj Sunderraman, Anu G. Bourgeois	2020	Proceedings of the 9th Computer Science Education Research Conference
Teamwork challenges and solution strategies of first-semester engineering students	A. Gorup, M. Grzunov, J. Petrovic et al.	2020	2020 43rd International Convention on Information, Communication and Electronic Technology (MIPRO)
The impact of collaborative group size on students' achievement and product quality in project-based learning environments	Ensaf Nasser Al Mulhim, Amany Ahmed Eldokhny	2020	International Journal of Emerging Technologies in Learning (iJET)

Continued on next page

Table 1 – continued from previous page

Title	Authors	Year	Venue
The impact of software team project measurements on students' performance in software engineering education	Bilal Al-Ahmad, Shadi Banitaan, Rami Alkhawaldeh	2020	Journal of Education and Practice
The impact of undergraduate mentorship on student satisfaction and engagement, teamwork performance, and team dysfunction in a software engineering group project	Claudia Iacob, Shamal Faily	2020	Proceedings of the 51st ACM Technical Symposium on Computer Science Education
Toward an improvement of engineering teaming skills through an In-house professionalism course	Mohammad Al-Abbas, Wissam Fawaz, Raymond Ghajar et al.	2020	IEEE Transactions on Education
Unreined students or not: modes of freedom in a project-based software engineering course	Oystein Nytro, Anh Nguyen-Duc, Hallvard Tratteberg et al.	2020	2020 IEEE 32nd Conference on Software Engineering Education and Training (CSEE&T)
Using continuous peer evaluation in team-based engineering capstone projects: a case study	Wilhelm A. Friess, Andrew J. Goupee	2020	IEEE Transactions on Education
Using GitHub in large software engineering classes. An exploratory case study	Miroslav Tushev, Grant Williams, Anas Mahmoud	2020	Computer Science Education
Web-based teamwork: distributed software development course under covid-19	Rui Zhang, Pedro Mestre, Carlos Serodio et al.	2020	2020 15th International Conference on Computer Science & Education (ICCSE)
Work in progress: fostering soft-skills of engineering students within scrum projects	Galyna Lutsenko, Gregory Lucenko	2020	2020 IEEE Global Engineering Education Conference (EDUCON)
Working with others and enjoying it: CS1 students' experience of small-group collaboration on a business simulation	Jyoti Bhardwaj	2020	2020 IEEE Frontiers in Education Conference (FIE)
A dev-ops practicum course for information technology majors	Adam Kaplan	2021	2021 International Conference on Computational Science and Computational Intelligence (CSCI)
A serious game for mobile phones used in a software engineering course: usability evaluation and educational effectiveness	Semira Maria Evangelou, Georgia Stamoulakatou, Michalis Xenos	2021	2021 IEEE Global Engineering Education Conference (EDUCON)
Active learning methodologies in STEM degrees jeopardized by COVID19	Alpha Pernia-Espinoza, Andres Sanz-Garcia, F. Javier Martinez-de-Pison-Ascacibar et al.	2021	2021 IEEE Global Engineering Education Conference (EDUCON)
An approach of project-based learning: bridging the gap between academia and industry needs in teaching integrated circuit design course	Xiaokun Yang	2021	IEEE Transactions on Education
An online educational hackathon to foster professional skills and intense collaboration on software engineering students	Caio Steglich, Sabrina Marczak, Luiz Guerra et al.	2021	Proceedings of the XXXV Brazilian Symposium on Software Engineering
Analysis-design-justification (ADJ): a framework to develop problem-solving skills	Ruben Acuña, Ajay Bansal	2021	2021 IEEE Global Engineering Education Conference (EDUCON)
ASEST framework: a proposal for improving teamwork by making cohesive software engineering student teams	Daymy Tamayo Avila, Wim Van Petegem, Arno Libotton	2021	European Journal of Engineering Education

Continued on next page

Table 1 – continued from previous page

Title	Authors	Year	Venue
Assessment of a hybrid software development process for student projects: a controlled experiment	Rafał Włodarski, Jean-Remy Falleri, Corinne Parvéry	2021	2021 IEEE/ACM 43rd International Conference on Software Engineering: Software Engineering Education and Training (ICSE-SEET)
Collaborative learning, self-efficacy, and student performance in CS1 POGIL	Aman Yadav, Chris Mayfield, Sukanya Kannan Moudgalya et al.	2021	Proceedings of the 52nd ACM Technical Symposium on Computer Science Education
Converting an undergrad-lab to an interactive E-learning experience that enables student teamwork	Ferdinand Keil, David Riehl, Klaus Hofmann	2021	2021 IEEE Frontiers in Education Conference (FIE)
Design and supervision model of group projects for active learning	Yi Meng Lau, Kyong Jin Shim, Swapna Gotipati	2021	2021 IEEE Frontiers in Education Conference (FIE)
Does self-efficacy correlate with positive emotion and academic performance in collaborative learning?	Nasrin Dehbozorgi, Mary Lou Maher, Mohsen Dorodchi	2021	2021 IEEE Frontiers in Education Conference (FIE)
Encouraging compiler optimization practice for undergraduate students through competition	Yu Zhang, Chunming Hu, Mingliang Zeng et al.	2021	Proceedings of the 26th ACM Conference on Innovation and Technology in Computer Science Education V. 1
Enhancing the student learning experience by adopting TDD and BDD in course projects	Fabio Gomes Rocha, Layse Santos Souza, Thiciane Suely Silva et al.	2021	2021 IEEE Global Engineering Education Conference (EDUCON)
Enhancing the student perception on software project management in computer science	Antonia Mas, Antoni-Lluís Mesquida, Ricardo Colomo-Palacio	2021	IEEE Transactions on Education
Experience of teaching a course on software engineering principles without a project	Paul W. McBurney, Christian Murphy	2021	Proceedings of the 52nd ACM Technical Symposium on Computer Science Education
Experiences and insights from using GitHub Classroom to support project-based courses	Maria Augusta Nelson, Lesandro Ponciano	2021	2021 Third International Workshop on Software Engineering Education for the Next Generation (SEENG)
Exploration of teammate trust and interaction dynamics in human-autonomy teaming	Mustafa Demir, Nathan J. McNeese, Jaime C. Gorman et al.	2021	IEEE Transactions on Human-Machine Systems
Improving soft skills in agile software development by team leader rotation	Jose Libreros, Ivan Viveros, Maria Trujillo et al.	2021	Proceedings of the Fifth International Congress on Information and Communication Technology
Improving student engagement with project-based learning: a case study in software engineering	Paula Morais, Maria Joao Ferreira, Bruno Veloso	2021	IEEE Revista Iberoamericana de Tecnologías del Aprendizaje
Investigating student insight in software engineering team projects:	Simona Motogna, Dan Suci, Arthur-Jozsef Molnar	2021	Proceedings of the 16th International Conference on Evaluation of Novel Approaches to Software Engineering
Learning from team quizzes in CS2	Yeajin Ham, Brandon Myers	2021	Proceedings of the 52nd ACM Technical Symposium on Computer Science Education
LEGO® serious play in software engineering education	Daniel Lopez-Fernandez, Aldo Gordillo, Fernando Ortega et al.	2021	IEEE Access

Continued on next page

Table 1 – continued from previous page

Title	Authors	Year	Venue
Metacognitive challenges to support self-reflection of students in online software engineering education	Daniela Pedrosa, Mario Madureira Fontes, Tania Araujo et al.	2021	2021 4th International Conference of the Portuguese Society for Engineering Education (CISPEE)
Online course on soft skills for global engineers	Ksenia N. Volchenkova, Elena A. Zhezhera	2021	2021 International Conference on Quality Management, Transport and Information Security, Information Technologies (IT&QM&IS)
Scaffolding student success in the wilds of open source contribution	Emily Lovell, James Davis	2021	2021 IEEE Frontiers in Education Conference (FIE)
SE-first: a new approach to software engineering education	Colin Maly, Suzette Person, Leen-Kiat Soh	2021	2021 IEEE Frontiers in Education Conference (FIE)
Soft skills development in computer science students via multinational and multidisciplinary GameDev project	Piotr Milczarski, Krzysztof Podlaski, Artur Hłobaż et al.	2021	Proceedings of the 52nd ACM Technical Symposium on Computer Science Education
Software development methodologies, HEIs, and the digital economy	Kawther Saeedi, Anna Visvizi	2021	Education Sciences
Software engineering project-based learning in an up-to-date technological context	Alejandro Adorjan, Martin Solari	2021	2021 IEEE URUCON
Student engagement in an online software engineering course	Bruce R. Maxim, Thomas Limbaugh, Jeffrey J. Yackley	2021	2021 IEEE Frontiers in Education Conference (FIE)
Student perception of software engineering factors supported by machine learning	Daniel Moritz Marutschke, Jeremy White, Uwe Serdult	2021	2021 International Conference on Electrical, Computer, Communications and Mechatronics Engineering (ICECCME)
Student perspectives on the purpose of peer evaluation during group game development projects	Alexander Mitchell, Terry Greer, Warwick New et al.	2021	Proceedings of the 2021 Conference on United Kingdom & Ireland Computing Education Research
Supporting real demands in software engineering with a four steps project-based learning approach	Leonardo Humberto Silva, Renata Xavier Castro, Marice Costa Guimaraes	2021	2021 IEEE/ACM 43rd International Conference on Software Engineering: Software Engineering Education and Training (ICSE-SEET)
Teaching software engineering using abstraction through modeling	Mohsen Dorodchi, Nasrin Dehbozorgi, Mohammadali Fallahian et al.	2021	Informatics in Education
Teaching the scrum master role using professional agile coaches and communities of practice	Maria Paasivaara	2021	2021 IEEE/ACM 43rd International Conference on Software Engineering: Software Engineering Education and Training (ICSE-SEET)
Teamwork with an automatic tutoring environment as learning strategy in programming courses	Armandina J. Leal-Flores, Luis H. Gonzalez-Guerra	2021	2021 IEEE Global Engineering Education Conference (EDUCON)
The development and validation of a framework for teaching software engineering through startup practice in higher education	Orges Cico, Anh Nguyen Duc, Letizia Jaccheri	2021	2021 IEEE Frontiers in Education Conference (FIE)

Continued on next page

Table 1 – continued from previous page

Title	Authors	Year	Venue
The impact of programming project milestones on procrastination, project outcomes, and course outcomes: a quasi-experimental study in a third-year data structures course	Clifford A. Shaffer, Ayaan M. Kazerouni	2021	Proceedings of the 52nd ACM Technical Symposium on Computer Science Education
The implementation of project-based learning approach in technical courses: an investigation into students' perceptions	Anh Thi Van Pham, Thien Huu Tran	2021	Proceedings of the 5th International Conference on Advances in Artificial Intelligence
Undergraduate In-class research experience for computer architecture students	Muhittin Yilmaz	2021	2021 International Conference on Computational Science and Computational Intelligence (CSCI)
Using Experiential Learning to Improve Student Attitude and Learning Quality in Software Engineering Education	Ferdinand Ndifor Che, Kenneth David Strang, Narasimha Rao Vajjhala	2021	International Journal of Innovative Teaching and Learning in Higher Education
Using git metrics to measure students' and teams' code contributions in software development projects	Sivana Hamer, Christian Quesada-López, Alexandra Martínez et al.	2021	CLEI Electronic Journal
Using jungian personality types for teaching teamwork in a software engineering capstone course	Vreda Pieterse, Sylvia Stuurman, Marko C.J.D. Van Eekelen	2021	Proceedings of the 52nd ACM Technical Symposium on Computer Science Education
Who does what? Work division and allocation strategies of computer science student teams	Anna Van Der Meulen, Efthimia Aivaloglou	2021	2021 IEEE/ACM 43rd International Conference on Software Engineering: Software Engineering Education and Training (ICSE-SEET)
WIP: engaging software engineering students in synchronous and asynchronous on-line course	Bruce Maxim, Thomas Limbaugh	2021	2021 ASEE Virtual Annual Conference Content Access Proceedings
A learner-centered technique for collectively configuring inputs for an algorithmic team formation tool	Emily M. Hastings, Sneha R. Krishna Kumaran, Karrie Karahalios et al.	2022	Proceedings of the 53rd ACM Technical Symposium on Computer Science Education - Volume 1
A longitudinal case study on the effects of an evidence-based software engineering training	Sebastián Pizard, Diego Vallespir, Barbara Kitchenham	2022	2022 IEEE/ACM 44th International Conference on Software Engineering: Software Engineering Education and Training (ICSE-SEET)
A three-year study on peer evaluation in a software engineering project course	Miguel Ehecatl Morales-Trujillo, Matthias Galster, Fabian Gilson et al.	2022	IEEE Transactions on Education
Adapting hackathon-honed skills toward software engineering capstone	Shawn Jordan, Cecilia La Place	2022	2022 ASEE Annual Conference & Exposition Proceedings
An eco-system approach to project-based learning in software engineering education	Daniel Stahl, Kristian Sandahl, Lena Bufoni	2022	IEEE Transactions on Education
Analyzing students' learning engagements using PLS-SEM: a case study in a small private online programming course	Bilegjargal Daramsenge, Nien-Lin Hsueh, Lien-Chi Lai	2022	Proceedings of the 2022 6th International Conference on Education and E-Learning
Building the professional competence of future programmers using methods and tools of flexible development of software applications	Olena G. Glazunova, Oleksandra V. Parkhomenko, Valentyna I. Korolchuk et al.	2022	Information Technologies and Learning Tools

Continued on next page

Table 1 – continued from previous page

Title	Authors	Year	Venue
Can pre-class github contributions predict success by student teams?	Jialin Cui, Ruochi Li, Kaida Lou et al.	2022	2022 IEEE/ACM 44th International Conference on Software Engineering: Software Engineering Education and Training (ICSE-SEET)
Collaborative learning through PBL and virtual monitoring	Sara Balderas-Diaz, Gabriel Guerrero-Contreras, M. Del Carmen De Castro-Cabrera	2022	2022 International Symposium on Computers in Education (SIIE)
Collaborative paradigm of teaching penetration testing using real-world university applications	Sufatrio, Jan Vykopal, Ee-Chien Chang	2022	Proceedings of the 24th Australasian Computing Education Conference
Collaborative project-based training and learning through multidisciplinary undergraduate research	Amar Khoukhi	2022	2022 IEEE Frontiers in Education Conference (FIE)
Comparing team evaluation software (team+ and CATME)	Amber Kemppainen, Amy Hamlin, Matthew Barron et al.	2022	2022 IEEE Frontiers in Education Conference (FIE)
Department-wide multi-semester community engaged learning initiative to overcome common barriers to service-learning implementation	Kathleen Timmerman, Michael Goldweber	2022	Proceedings of the 53rd ACM Technical Symposium on Computer Science Education
Developing 21st century skills with project-based learning: an experience report in the introductory course of software engineering	Sujitra Arwatchananukul, Pattariya Singpant, Nacha Chondamrongkul et al.	2022	2022 Joint International Conference on Digital Arts, Media and Technology with ECTI Northern Section Conference on Electrical, Electronics, Computer and Telecommunications Engineering (ECTI DAMT & NCON)
Do teamwork guidelines improve peer assessment accuracy or attitudes during collaborative learning?	Meixiu Lu, Ming Ming Chiu	2022	IEEE Transactions on Education
Educating reflective systems developers at scale: towards “productive feedback” in a semi-capstone large-scale software engineering course	Torgeir Dingsoyr	2022	2022 IEEE Frontiers in Education Conference (FIE)
Engagement contexts of software engineering education projects	E. Moira Watson, David Cutting	2022	2022 31st Annual Conference of the European Association for Education in Electrical and Information Engineering (EAEEIE)
Enhancing the quality and student attitude on capstone projects through activity-based learning and reduced social loafing	Uthpala Samarakoon, Kalpani Manathunga, Asanthika Imbulpitiya	2022	Computer supported education
Exploring student challenges in an online project-based course	Simona Motogna, Dan Mircea Suciu, Arthur-Jozsef Molnar	2022	2022 IEEE/ACM First International Workshop on Designing and Running Project-Based Courses in Software Engineering Education (DREE)
From undergraduate (software) capstone projects to start-ups: challenges and opportunities in higher institutions of learning	Daniel Ogenrwot, Geoffrey Olok Tabo, Kevin Aber et al.	2022	Proceedings of the Federated Africa and Middle East Conference on Software Engineering

Continued on next page

Table 1 – continued from previous page

Title	Authors	Year	Venue
Gender influence on communication initiated within student teams	Rita Garcia, Chieh-Ju Liao, Ariane Pearce et al.	2022	Proceedings of the 53rd ACM Technical Symposium on Computer Science Education - Volume 1
Improving soft skills based on students' traces in problem-based learning environments	Houda Tadjer, Yacine Lafifi, Hassina Seridi-Bouchelaghem et al.	2022	Interactive Learning Environments
Improving teamwork in agile software engineering education: the ASEST+ framework	Daymy Tamayo Avila, Wim Van Petegem, Monique Snoeck	2022	IEEE Transactions on Education
Incorporating global learning perspectives in a freshman computing curriculum	Shamima Mithun, Nancy Goldfarb	2022	2022 IEEE Frontiers in Education Conference (FIE)
Information technologies as a tool for bachelor's engineering teamwork skills development	Anna M. Markus, Lada N. Ovinova, Elena G. Shraiber et al.	2022	2022 International Conference on Quality Management, Transport and Information Security, Information Technologies (IT&QM&IS)
Integrating hackathons into an online cybersecurity course	Abasi-amefon Obot Affia, Alexander Nolte, Raimundas Matulevičius	2022	2022 IEEE/ACM 44th International Conference on Software Engineering: Software Engineering Education and Training (ICSE-SEET)
Integrating hackathons into an online cybersecurity course	Abasi-Ame fon Obot Affia, Alexander Nolte, Raimundas Matulevicius	2022	2022 IEEE/ACM 44th International Conference on Software Engineering: Software Engineering Education and Training (ICSE-SEET)
Learning communication skills in software development management courses: an active learning experience	Carlos Fontela, Gonzalo Damian Santos	2022	2022 IEEE Biennial Congress of Argentina (ARGENCON)
Learning CS subjects of professional software development and team projects	Yu Chyi Chan, Chian Min Gan, Chun Yu Lim et al.	2022	2022 IEEE International Conference on Teaching, Assessment and Learning for Engineering (TALE)
Learning experiences and analysis in professional software development subject	Madeline Jia Qi Thong, Arico Qi Xian Yau, Sida Andy Goh et al.	2022	2022 IEEE International Conference on Teaching, Assessment and Learning for Engineering (TALE)
Learning soft skills through distributed software development	Emily Laue Christensen, Maria Paasivaara	2022	Proceedings of the International Conference on Software and System Processes and International Conference on Global Software Engineering
On competence and virtue in engineering and computing education	Stephen Frezza, Mani Mina	2022	2022 IEEE Frontiers in Education Conference (FIE)
Operationalizing team commitment in a project-based learning environment	Aparajita Jaiswal, Tugba Karabiyik, Paul Thomas et al.	2022	2022 IEEE Frontiers in Education Conference (FIE)
Optimize matching for project team members in practice course based on extenics	Ruoyan Wei, Xiaojuan Li, Weijie Wang et al.	2022	2022 3rd International Conference on Education, Knowledge and Information Management (ICEKIM)

Continued on next page

Table 1 – continued from previous page

Title	Authors	Year	Venue
Perceived scrum values, conflict resolution ability, and cultural self-awareness among system design and analysis students	A. Jaiswal, K. Patel, D. Patel et al.	2022	ASEE Annual Conference proceedings
Piloting the learning by development action model pedagogy in UK HEIs	Taina Lintilä, Mark Zarb	2022	2022 IEEE Frontiers in Education Conference (FIE)
Refining a risk framework for student group projects	Diana Kirk, Andrew Luxton-Reilly, Ewan Tempero	2022	Proceedings of the 22nd Koli Calling International Conference on Computing Education Research
Reform of active pedagogy in the age of covid	Faten Ziadi, Naouel Boughattas, Wissal Neji	2022	2022 IEEE Global Engineering Education Conference (EDUCON)
SCRUM applied to problem-based learning: a hybrid model for managing the teaching-learning process	Sam Da Silva Devincenzi, Viviani Rios Kwecko, Alessandro De Lima Bicho et al.	2022	2022 IEEE Frontiers in Education Conference (FIE)
Software engineering, bridging theory and practice in an agile learning environment	Dena Hussain, Linda Söderlindh	2022	2022 IEEE Global Engineering Education Conference (EDUCON)
Student engagement in software engineering group projects: an action research study	Yue Li, Soon Phei Tin, Charlie Reis	2022	Proceedings of the 2022 5th International Conference on Education Technology Management
Student-sponsored projects in a capstone course: reflections and lessons learned	Ethan Bütt, Suzette Person, Christopher Bohn	2022	Proceedings of the ACM/IEEE 44th International Conference on Software Engineering: Software Engineering Education and Training
Student-sponsored projects in a capstone course: reflections and lessons learned	Ethan Butt, Suzette Person, Christopher Bohn	2022	2022 IEEE/ACM 44th International Conference on Software Engineering: Software Engineering Education and Training (ICSE-SEET)
Students' feedback about their experiences in EPICS using natural language processing	Isil Anakok, Johnny Woods, Mark Huerta et al.	2022	2022 IEEE Frontiers in Education Conference (FIE)
Supporting peripheral perception in distributed teams by enforced exposure to chat messages	Miroslav Novotny, Valentino Vranic	2022	IEEE Transactions on Human-Machine Systems
Teaching agile in an agile way: a case from the first iteration in a university	Necmettin Ozkan, Ozden Ozcan-Top, Seval Bal et al.	2022	2022 3rd International Informatics and Software Engineering Conference (IISEC)
Teaching agile operation and leadership through linked university courses	Enikő Ilyés	2022	Teaching Mathematics and Computer Science
Teaching software engineering as programming over time	Markus Hofbauer, Christoph Bachhuber, Christopher Kuhn et al.	2022	Proceedings of the 4th International Workshop on Software Engineering Education for the Next Generation
Team building elements and their importance according to STEAM students	Karla Krasnic, Petra Zigman, Tihana Babic	2022	2022 45th Jubilee International Convention on Information, Communication and Electronic Technology (MIPRO)
Team formation on the basis of belbin's roles to enhance students' performance in project based learning	A. Aranzabal, E. Epelde, M. Artetxe	2022	Education for Chemical Engineers

Continued on next page

Table 1 – continued from previous page

Title	Authors	Year	Venue
Team harmony before, during, and after COVID-19	Noa Heyl, Elisa Baniassad, Oluwakemi Ola	2022	Proceedings of the 2022 ACM SIGPLAN International Symposium on SPLASH-E
The impact and measurement of today's learning technologies in teaching software engineering course using design-based learning and project-based learning	Chetna Gupta	2022	IEEE Transactions on Education
Understanding students' software development projects: effort, performance, satisfaction, skills and their relation to the adequacy of outcomes developed	Sherlock A. Licorish, Matthias Galster, Georgia M. Kapitsaki et al.	2022	Journal of Systems and Software
Using design-based research to layer career-like experiences onto software development courses	Christine Bakke, Rena Sakai	2022	Journal of Information Technology Education: Innovations in Practice
Using process mining to understand students' and teams' dynamics	Pasquale Ardimento, Mario Luca Bernardi, Marta Cimitile	2022	Higher education learning methodologies and technologies online
What makes team[s] work? A study of team characteristics in software engineering projects	Kai Presler-Marshall, Sarah Heckman, Kathryn T. Stolee	2022	Proceedings of the 2022 ACM Conference on International Computing Education Research - Volume 1
Who engaged in the team-based assessment? Leveraging EdTech for a self and intra-team peer-assessment solution to free-riding	Tiffany K. Gunning, Xavier A. Conlan, Paul K. Collins et al.	2022	International Journal of Educational Technology in Higher Education
20 years of EPICS at butler university: experiences and lessons learned	Lucas Johnson, Panagiotis Linos	2023	2023 IEEE Frontiers in Education Conference (FIE)
Advancing CyberSecurity education and training: practical case study of running capture the flag (CTF) on the metaverse vs. Physical settings	Jamal N. Al-Karaki, Marwan Omar, Amjad Gawanmeh et al.	2023	2023 International Conference on Intelligent Metaverse Technologies & Applications (iMETA)
Affective computing: a topic-based SER approach on collaborative discussions in academic setting	Nasrin Dehbozorgi, Mourya Teja Kunuku	2023	2023 IEEE Frontiers in Education Conference (FIE)
An AI-based approach for grading students' collaboration	Bojan B. Tomić, Anisja D. Kijevčanin, Zoran V. Ševarec et al.	2023	IEEE Transactions on Learning Technologies
An authentic peer-led team learning program for community colleges: a recruitment, retention, and completion instrument for face-to-face and online modality	Christian Servin, Myshie Pagel, Ernest Webb	2023	Proceedings of the 54th ACM Technical Symposium on Computer Science Education V. 1
An optimal grouping and regrouping method for effective collaborative learning: leveraging the group dynamics	Soheila Garshasbi, Sabine Graf, Mahmood Asgari et al.	2023	2023 IEEE International Conference on Teaching, Assessment and Learning for Engineering (TALE)
Analysis of student pair teamwork using GitHub activities	Niki Gitinabard, Zhikai Gao, Sarah Heckman et al.	2023	Journal of Educational Data Mining
Assessing the impact of curriculum integration on the intercultural learning gains of first-year computer science students	Aparajita Jaiswal, Laura Starr, Aletha Stahl	2023	2023 IEEE Frontiers in Education Conference (FIE)

Continued on next page

Table 1 – continued from previous page

Title	Authors	Year	Venue
Categorizing projects for software engineering capstone courses	Attika Rehman, Monica Divitini	2023	Norsk IKT-konferanse for forskning og utdanning
Combining GitHub, chat, and peer evaluation data to assess individual contributions to team software development projects	Christopher Hundhausen, Phill Conrad, Olusola Adesope et al.	2023	ACM Transactions on Computing Education
Complete integration of team project-based learning into a database syllabus	Sergio Iserte, Vicente R. Tomás, Miguel Pérez et al.	2023	IEEE Transactions on Education
Cooperative learning and Co-regulation: exploring students' teamwork strategies in higher education	Sakhi Aggrawal, Jorge A. Cristancho, Devang A. Patel et al.	2023	2023 IEEE Frontiers in Education Conference (FIE)
Coordination and awareness mechanisms to support collaboration in project-based software engineering courses	Maira Marques, Roc Meseguer, Sergio F. Ochoa et al.	2023	2023 26th International Conference on Computer Supported Cooperative Work in Design (CSCWD)
Designing an overseas experiential course in data science	Hua Leong Fwa, Graham Ng	2023	2023 IEEE International Conference on Teaching, Assessment and Learning for Engineering (TALE)
Diversity and teamwork in student software teams	Isabella Graßl, Stephan Krusche, Gordon Fraser	2023	Proceedings of the 5th European Conference on Software Engineering Education
Effectiveness of video-based training for face-to-face communication skills of software engineers: evidence from a three-year study	Antonija Mitrovic, Matthias Galster, Sanna Malinen et al.	2023	ACM Transactions on Computing Education
Engaged student learning through IoT-based capstone projects: particular look at student engagement in historically underrepresented groups	Mehrube Mehrubeoglu, Lifford McLaughlan, David Hicks et al.	2023	2023 IEEE Frontiers in Education Conference (FIE)
Enhancing design thinking in engineering students with project-based learning	Cuiling Jiang, Yilin Pang	2023	Computer Applications in Engineering Education
Experiences and lessons learned from real-world projects in software engineering subject	Yan Hern Ryan Sim, Zhi Zhan Lua, Kahbailan Kalisavvam Kelaver et al.	2023	2023 IEEE 35th International Conference on Software Engineering Education and Training (CSEE&T)
Explicitly characterizing team structures in teaching team-based project courses	Mingxian Jin	2023	J. Comput. Sci. Coll.
Exploring agile methodologies in educational software development- a comparative analysis and project management insights	Vandana Kate, Shilpa Bhalerao, Vanshika Sharma	2023	2023 IEEE International Conference on ICT in Business Industry & Government (ICTBIG)
Exposing software engineering students to stressful projects: does diversity matter?	Isabella Graßl, Gordon Fraser, Stefan Trieflinger et al.	2023	2023 IEEE/ACM 45th International Conference on Software Engineering: Software Engineering Education and Training (ICSE-SEET)
How ChatGPT will change software engineering education	Marian Daun, Jennifer Brings	2023	Proceedings of the 2023 Conference on Innovation and Technology in Computer Science Education V. 1

Continued on next page

Table 1 – continued from previous page

Title	Authors	Year	Venue
Implementation of PjBL with remote lab enhances the professional skills of engineering students	V. Hayashi, J. Dutra, F. Almeida et al.	2023	IEEE Transactions on Education
Improving accuracy of self-and-peer assessment in engineering technology capstone	Frederick C. Berry, Wanju Huang, Marisa Exter	2023	IEEE Transactions on Education
Improving soft skills through an interdisciplinary approach in a realistic context between education and CS students in an HCI course	Maximiliano Paredes-Velasco, Mónica Arnal-Palacián, Jaime Urquiza-Fuentes et al.	2023	IEEE Transactions on Education
Improving teamwork in software engineering projects in higher education	Ser Wee Darren Quek, Jared Kai Fu Teo, Nasruddine Louahemmsabah et al.	2023	2023 IEEE Frontiers in Education Conference (FIE)
Improving the methodology of teaching specialized subjects in the preparation of future computer engineering on the basis of innovative technologies	Atamuratov Rasuljon Kadirjanovich, Abdullayeva Nazokat Isayevna, Primova Gozal Gulomjonovna et al.	2023	2023 8th International Conference on Computer Science and Engineering (UBMK)
Information system development by using agile teamwork and service-learning	Vannessa Duarte, Margareth Cleveland-Slimming, Cristian Vidal et al.	2023	IEEE Transactions on Education
Infrastructural elements to fostering educational equity in engineering	Katrina Robertson, Hadi Ali, Jonathan Adams	2023	2023 IEEE Frontiers in Education Conference (FIE)
Investigating reflection in undergraduate software development teams: an analysis of online chat transcripts	Christopher Hundhausen, Phill Conrad, Olusola Adesope et al.	2023	Proceedings of the 54th ACM Technical Symposium on Computer Science Education V. 1
Learning emergent digital technologies: the experience in the internet of things course unit	Paulo Leitão, Luis Piardi, Lucas Sakurada et al.	2023	2023 IEEE 21st International Conference on Industrial Informatics (INDIN)
Leveraging diversity in software engineering education through community engaged learning and a supportive network	Nowshin Nawar Arony, Kezia Devathanan, Ze Shi Li et al.	2023	2023 IEEE/ACM 45th International Conference on Software Engineering: Software Engineering Education and Training (ICSE-SEET)
Model and methodology for developing empathy: an experience in computer science engineering	Cecilia Sanz, Teresa Coma-Roselló, Antonio Aguelo et al.	2023	IEEE Transactions on Education
Modeling team interaction and decision-making in agile human-machine teams: quantum and dynamical systems perspective	Mustafa Demir, Mustafa Canan, Myke C. Cohen	2023	IEEE Transactions on Human-Machine Systems
On the impact of grading on teamwork quality in a software engineering capstone course	María Cecilia Bastarrica, Francisco J. Gutierrez, María Marques et al.	2023	IEEE Access
Planning for difference: preparing students to create flexible and elaborated team charters that can adapt to support diverse teams	Maria Feuer, Joanna Wolfe	2023	IEEE Transactions on Professional Communication
Relationships between social interactions and Belbin role types in collaborative agile teams	Diego Monsalves, Héctor Cornide-Reyes, Fabián Riquelme	2023	IEEE Access
Shooting for the stars: how a STEM initiative has evolved to address gender challenges in work and education	Marie Travers, Jenna Bromell, Katie Crowley et al.	2023	2023 IEEE/ACM 4th Workshop on Gender Equity, Diversity, and Inclusion in Software Engineering (GEICSE)

Continued on next page

Table 1 – continued from previous page

Title	Authors	Year	Venue
Software startup within a university – producing industry-ready graduates	Saara Tenhunen, Tomi Männistö, Petri Ihantola et al.	2023	2023 IEEE/ACM 45th International Conference on Software Engineering: Software Engineering Education and Training (ICSE-SEET)
Structured software development versus agile software development: a comparative analysis	Alok Mishra, Yehia Ibrahim Alzoubi	2023	International Journal of System Assurance Engineering and Management
Student autonomy in collaborative learning: effects of meeting time and team consistency	Hongxuan Chen, Morgan M. Fong, Geoffrey L. Herman et al.	2023	2023 IEEE Frontiers in Education Conference (FIE)
Student perceptions of teamwork, conflict, and industry preparedness in engineering interdisciplinary capstone design	Lisa Schibeli, Olivia Ryan, Susan Sajadi	2023	2023 IEEE Frontiers in Education Conference (FIE)
Student reflections on service-learning in software engineering and their experiences with non-technical clients	Stan Kurkovsky	2023	Proceedings of the ACM Conference on Global Computing Education Vol 1
Students' perceptions of integrating a contribution measurement tool in software engineering projects	Sivana Hamer, Christian Quesada-López, Marcelo Jenkins	2023	2023 IEEE 35th International Conference on Software Engineering Education and Training (CSEE&T)
Supporting online collaborative work at scale: a mixed-methods study of a learning analytics tool	Xavier Ochoa, Vanessa Echeverria, Gladys Carrillo et al.	2023	Proceedings of the Tenth ACM Conference on Learning @ Scale
Teaching of web design and programming as a role-playing team building game	Bulgaria The Department Of Education In Mathematics, Ivelina V. Velcheva, Hristo Hristov	2023	International Journal of Information and Education Technology
Teamwork dynamics in the context of large-size software development courses	Alejandra J. Magana, Theodora Amuah, Sakhi Aggrawal et al.	2023	International Journal of STEM Education
Teamwork improvement in group projects of professional software development	Esther Su Wee Tan, Muhammad Ridhwan Bin Sani, Nor Atiqah Shahira B. Norshahid et al.	2023	2023 3rd International Conference on Educational Technology (ICET)
Teamwork in CS1: student learning and experience with POGIL	Helen H. Hu, Aman Yadav, Donna M. Gavin et al.	2023	Proceedings of the 54th ACM Technical Symposium on Computer Science Education V. 1
The daily smirk: a preliminary prototype for continuous peer assessment of team-based projects	Yu Lu, André Van Der Hoek	2023	2023 IEEE 35th International Conference on Software Engineering Education and Training (CSEE&T)
The impact of unequal contributions in student software engineering team projects	Kamilla Kopec-Harding, Sukru Eraslan, Bowen Cai et al.	2023	Journal of Systems and Software
Training students to choose their agile practices and tools	Paolo Ciancarini, Andrea Loretti, Marcello Missiroli et al.	2023	2023 IEEE 35th International Conference on Software Engineering Education and Training (CSEE&T)
Transformative pedagogy as a reflective approach for promoting intercultural self-awareness in the context of teamwork	Irene Hensista, Shreya Guddeti, Devang A. Patel et al.	2023	2023 IEEE Frontiers in Education Conference (FIE)

Continued on next page

Table 1 – continued from previous page

Title	Authors	Year	Venue
Trends and challenges of project-based learning in computer science and engineering education	Sadaqat Ur Rehman	2023	The 15th International Conference on Education Technology and Computers
Understanding teamwork in dynamic contexts: an study in a software engineering project course	Sergio F. Ochoa, Maira Marques Samary, Luis Silvestre et al.	2023	2023 International Conference on Computational Science and Computational Intelligence (CSCI)
Understanding the teamwork challenges of software engineering students	L. Dorić, N. Luburić, J. Slivka et al.	2023	2023 46th MIPRO ICT and Electronics Convention (MIPRO)
Use of business games in the learning process	Janis Amolins, Sabina Katalnikova, Natalya Prokofyeva et al.	2023	2023 IEEE 64th International Scientific Conference on Information Technology and Management Science of Riga Technical University (ITMS)
Using automatic program assessment in a software development project course	Uwe M. Borghoff, Mark Minas, Kim Mönch	2023	Proceedings of the 5th European Conference on Software Engineering Education
Using IF-AT cards to engage students in deeper learning of course content	Ana Luna, Cruz Izu	2023	IEEE Revista Iberoamericana de Tecnologías del Aprendizaje
Using PBL and agile to teach artificial intelligence to undergraduate computing students	Vitor Augusto Menten De Barros, Henrique Mohallem Paiva, Victor Takashi Hayashi	2023	IEEE Access
What makes project teams succeed? Students' post-COVID perceptions on IT project management education fostering professional skills	Dominik Dolezal, Renate Motschnig	2023	2023 IEEE Frontiers in Education Conference (FIE)
“Software is the easy part of software engineering” - lessons and experiences from a large-scale, multi-team capstone course	Ze Shi Li, Nowshin Nawar Arony, Kezia Devathasan et al.	2023	2023 IEEE/ACM 45th International Conference on Software Engineering: Software Engineering Education and Training (ICSE-SEET)
C4 skills in the engineering graduate: a study to align software engineering education with market-driven software industry needs	Chetna Gupta, Varun Gupta	2024	IEEE Transactions on Education
AI-based automatic detection of online teamwork engagement in higher education	Alejandra J. Magana, Syed Tanzim Mubarrat, Dominic Kao et al.	2024	IEEE Transactions on Learning Technologies
An empirical study of the content and quality of sprint retrospectives in undergraduate team software projects	Christopher Hundhausen, Phillip Conrad, Ahsun Tariq et al.	2024	2024 IEEE/ACM 46th International Conference on Software Engineering: Software Engineering Education and Training (ICSE-SEET)
Antecedents of student team formation in higher education	Carlos A. De Matos Fernandes, Marion Hoffman, Jasperina Brouwer	2024	Learning and Instruction
Assessing soft skills development in informatics students through project-based learning and agile frameworks	Vanessa Maribel Choque-Soto, Victor Dario Sosa-Jauregui	2024	2024 International Symposium on Accreditation of Engineering and Computing Education (ICACIT)
Assessing student engagement, success, leadership and teamwork skills with respect to team role selection and execution	Edward Latorre	2024	2024 ASEE Annual Conference & Exposition Proceedings

Continued on next page

Table 1 – continued from previous page

Title	Authors	Year	Venue
ASSESSING the implementation of a unified software project across various courses in higher education	Leah Velasco Barbaso, Eugene C Busico, Ralph P Laviste et al.	2024	Proceedings of the 2024 8th International Conference on Digital Technology in Education (ICDTE)
Bridging theory to practice in software testing teaching through team-based learning (TBL) and open source software (OSS) contribution	Elaine Venson, Reem Alfayez	2024	Proceedings of the 46th International Conference on Software Engineering: Software Engineering Education and Training
Building bridges: a holistic approach to web development education through community engaged learning and social justice initiatives	Amalia Rusu, Adrian Rusu, Aarushi Vijay	2024	2024 IEEE Frontiers in Education Conference (FIE)
Characterizing team cognition within software engineering teams in an undergraduate course	Alejandra J. Magana, Aparajita Jaiswal, Theodora Loyce Amuah et al.	2024	IEEE Transactions on Education
Code collaborate: dissecting team dynamics in first-semester programming students	Santiago Berrezueta-Guzman, Patrick Bassner, Stefan Wagner et al.	2024	2024 21st International Conference on Information Technology Based Higher Education and Training (ITHET)
Developing future computational thinking in foundational CS education: a case study from a liberal education university in India	Balaji Kalluri, Prajish Prasad, Prakrati Sharma et al.	2024	IEEE Transactions on Education
Educator experiences of low overhead student project risk management	Diana Kirk, Andrew Luxton-Reilly, Ewan Tempero et al.	2024	Proceedings of the 26th Australasian Computing Education Conference
Effectiveness of implementing team contracts in a client-sponsored project-based learning course	Arko Barman	2024	2024 IEEE Frontiers in Education Conference (FIE)
Engineering better communicators via authentic integration and communicative self-efficacy	Allison Hutchison	2024	2024 47th MIPRO ICT and Electronics Convention (MIPRO)
Enhancing computer science education: gamification for developing technical and transversal competences	Simena Dinas, Carlos Giovanni Hidalgo, Fabian H. Posso Gordillo et al.	2024	2024 IEEE VII Congreso Internacional en Inteligencia Ambiental, Ingeniería de Software y Salud Electrónica y Móvil (AmITIC)
Enhancing CS education with LAs using AI-empowered AIELA program	Vijayalakshmi Ramasamy, Eli Kulpinski, Thomas Beaupre et al.	2024	2024 IEEE Frontiers in Education Conference (FIE)
Enhancing engineering education through project-based learning: a case study of biometry and computer vision courses	António J. R. Neves, Fabianne Ribeiro	2024	2024 IEEE Global Engineering Education Conference (EDUCON)
Enhancing learning experiences in erasmus blended intensive programs	Laura Grindei, Rodica Holonec, Sara Blanc Clavero et al.	2024	2024 21st International Conference on Information Technology Based Higher Education and Training (ITHET)
Evaluating individual contributions in teams for project-based learning in data science	Arko Barman, Su Chen, Genevera I. Allen	2024	2024 IEEE Frontiers in Education Conference (FIE)
Evaluating the role of transdisciplinary course design and its efficacy in teaching a personal communication course for engineering students	Brainerd Prince, Rukmani Keshav, Anenya Kez	2024	2024 IEEE Frontiers in Education Conference (FIE)
Exploring peer evaluation methods in group projects of software engineering education	Mohammad Rizqullah Hafizh Bin Mohamed Riduwan, Mark Peng Jung Zen, Oh Jia Wei Darien et al.	2024	2024 IEEE Frontiers in Education Conference (FIE)

Continued on next page

Table 1 – continued from previous page

Title	Authors	Year	Venue
Factors influencing the acquisition of soft skills in a collaborative learning environment supported by game-based application	Jose Amelio Medina-Merodio, Ana Castillo-Martinez, Roberto Barchino et al.	2024	IEEE Access
Fostering agile IT project management and interpersonal skills using AI-enhanced game-based learning	Dominik Dolezal, Yllka Velaj, Lukas Spreitzer et al.	2024	2024 IEEE Frontiers in Education Conference (FIE)
From traditional to agile methodologies in software project management education: a case study	Mickael Fonseca Santos, Ricardo Angelo Filipe, Joao Carlos Cunha	2024	Proceedings of the 2024 16th International Conference on Education Technology and Computers
GA2Graph: a data-driven approach to visualizing and analyzing collaborative learning	Abdussalam Alawini, Isa Hajara-Yasmin, Jiabao Xu et al.	2024	2024 IEEE Frontiers in Education Conference (FIE)
Guess the learning type: a micro-designed gamified approach to machine learning types	Nur Zareen Zulkarnain, Sharifah Sakinah Syed Ahmad	2024	2024 International Conference on TVET Excellence & Development (ICTeD)
Hands-on experiential learning of machine learning concepts through senior design projects: a case study	Ahmed A. Hussain, Samir El-Nakla, Sadiq Al Huwaidi et al.	2024	2024 6th International Symposium on Advanced Electrical and Communication Technologies (ISAECT)
Improving software engineering teamwork with structured feedback	Victor Weiqi Huang, Kori Krueger, Taya Cohen et al.	2024	Proceedings of the 55th ACM Technical Symposium on Computer Science Education V. 1
Innovative approach to agile education: generative AI-supported planning poker simulation	Khalid Nafil, Youssef Lefdaoui	2024	2024 21st International Conference on Information Technology Based Higher Education and Training (ITHET)
Innovative practice: agile training for year-long capstone project	William Eberle, Gerald C. Gannod, Eric Brown	2024	2024 IEEE Frontiers in Education Conference (FIE)
Integrating international research-innovation projects and working life partners into cybersecurity degree program	Jyri Rajamäki, Paresh Rathod, Pasi Kämppi	2024	2024 IEEE Global Engineering Education Conference (EDUCON)
Integrating usability into software engineering course projects	Nihal Mezi Cetin	2024	Journal of Computer and Education Research
Investigating computer science students perceptions of team coaching	Vanessa Tudor, Dominik Dolezal, Renate Motschnig	2024	2024 IEEE Frontiers in Education Conference (FIE)
Managing students in hybrid software project classes	Bruce R. Maxim, Belen A. Garcia	2024	2024 IEEE Frontiers in Education Conference (FIE)
Pay it forward: use of a course-based discussion platform to deepen content and professional knowledge in an engineering project course	Edward Latorre-Navarro	2024	2024 IEEE Frontiers in Education Conference (FIE)
Providing technical support to sustain student motivation and engagement in software engineering project-based learning	Ahmad D. Suleiman, David Shepherd, Jan DeWaters et al.	2024	2024 IEEE Frontiers in Education Conference (FIE)
Relating team atmosphere and group dynamics to student software development teams' performance	Sherlock A. Licorish, Daniel Alencar Da Costa, Elijah Zolduoarrati et al.	2024	Information and Software Technology
Strategies and implications of peer assessment in software engineering education	Joy Yee Shing Cheng, Zi Jian Adrian Pang, Elias Isaac Huai-En Lim et al.	2024	2024 IEEE Frontiers in Education Conference (FIE)

Continued on next page

Table 1 – continued from previous page

Title	Authors	Year	Venue
Students' perceptions on adaptations and workload versus artifacts in a project-based scrum-variant course	Georgia M. Kapitsaki	2024	IEEE Transactions on Education
Teaching robotic concepts using project-based learning: a case study	Bianca Cassemiro Lima, Henrique Mohallem Paiva, Marcos R. O. A. Maximo	2024	2024 IEEE World Engineering Education Conference (EDUNINE)
Teaching software development for real-world problems using a microservice-based collaborative problem-solving approach	Yi Meng Lau, Christian Michael Koh, Lingxiao Jiang	2024	2024 IEEE/ACM 46th International Conference on Software Engineering: Software Engineering Education and Training (ICSE-SEET)
Teamwork conflict management training and conflict resolution practice via large language models	Sakhi Aggrawal, Alejandra J. Magana	2024	Future Internet
The application of the teaching model of “promoting learning through competition, promoting teaching through competition” in the course of data analysis and mining	Zhaojing Wang, Li Li, Bangchao Wang et al.	2024	2024 IEEE 24th International Conference on Software Quality, Reliability, and Security Companion (QRS-C)
The impact of the national cyber league NCL on students skills in cybersecurity	Yasser Alshehri	2024	2024 IEEE Frontiers in Education Conference (FIE)
The PBL methodology in a hybrid class in an integrative systems engineering course	Elena Sonia Paula Espinoza Ríos, Felipe Néstor Gutarra Meza, Isabel C. Barragan Arias et al.	2024	2024 International Symposium on Accreditation of Engineering and Computing Education (ICACIT)
The progression of competencies and dispositions: how do maturing software engineers compare?	Tony Clear, Alison Clear	2024	2024 IEEE Frontiers in Education Conference (FIE)
The role of team composition in agile software development education: a gendered perspective	Gyda Elisa Sæter, Viktoria Stray, Steffen Almås et al.	2024	Agile processes in software engineering and extreme programming
Transdisciplinarity and team-based learning: strategies for an introductory programming course	Rocio Ramos-Rodriguez, Maria Calle, Garis Coronell et al.	2024	IEEE Transactions on Education
Transforming the client relationship to support large capstone classes	Bowen Hui, Samantha Hodge, Dilpreet Samra	2024	2024 IEEE Frontiers in Education Conference (FIE)
Using ChatGPT in undergraduate computer science and software engineering courses: a students' perspective	Noah Andersen-Kiel, Panagiotis Panos Linos	2024	2024 IEEE Frontiers in Education Conference (FIE)
Using GitHub analytics to assess the quality of collaboration in software engineering teams	Quan Le, Kiet Phan, Bowen Hui et al.	2024	2024 IEEE Frontiers in Education Conference (FIE)
Using vignettes to categorize behaviors that students associate with dispositions	Mihaela Sabin, Renée McCauley, Bonnie MacKellar et al.	2024	2024 IEEE Frontiers in Education Conference (FIE)
WIP: a systematic approach to screen and align service-learning projects for optimal student outcomes	Chad A. Williams, Stan Kurkovsky, Mikey Goldweber et al.	2024	2024 IEEE Frontiers in Education Conference (FIE)
WIP: ChatVis: enhancing academic team collaboration through WhatsApp chat analytics	Héctor G. Pérez-González, Raymundo A. González-Grimaldo, Joshua Chibuike Nwokeji et al.	2024	2024 IEEE Frontiers in Education Conference (FIE)
WIP: industry 4.0 robotics - an interdisciplinary approach to deep learning	Chad A. Williams, Haoyu Wang, Stan Kurkovsky et al.	2024	2024 IEEE Frontiers in Education Conference (FIE)

Continued on next page

Table 1 – continued from previous page

Title	Authors	Year	Venue
Working with large code bases: a cognitive apprenticeship approach to teaching software engineering	Anshul Shah, Jerry Yu, Thanh Tong et al.	2024	Proceedings of the 55th ACM Technical Symposium on Computer Science Education V. 1
“Think before you scrum” - an experience report on daily scrums to aid reflection	Matthew Minish, Matthias Galster, Fabian Gilson	2024	2024 36th International Conference on Software Engineering Education and Training (CSEE&T)
A simulation-based approach to enhance conflict resolution through non-violent communication and large language models in teamwork	Miguel A. Feijoo-Garcia, Jorge A. Crisancho, James Dworkin et al.	2025	2025 IEEE Frontiers in Education Conference (FIE)
Active learning methodology for advanced agile software engineering and DevOps practices	Juan Ignacio Irabedra, Martin Solari	2025	2025 IEEE Engineering Education World Conference (EDUNINE)
Agile methodologies in education: the common ground of personalized learning and project-based learning	Nuno Pombo, Carlos Cunha	2025	2025 IEEE/ACIS 23rd International Conference on Software Engineering Research, Management and Applications (SERA)
Agile social software engineer: an empirical approach to mitigate and manage social debts in agile environments	Mohammed Nazel Alshammari, Wendy Yáñez-Pazmino, Carlos Mera-Gómez et al.	2025	ACM Transactions on Computing Education
AI literacy through a project-based learning course	Marco Ho, Carly Orr, Rebecca Jeon et al.	2025	2025 IEEE Smart World Congress (SWC)
An enhanced framework for sustainable education using project-based learning	Sayedra Rahnuma Akthar, Mahady Hasan, Awindrela Roy Dristy et al.	2025	2025 IEEE/ACIS 23rd International Conference on Software Engineering Research, Management and Applications (SERA)
An exploration of how generative AI affects workflow and collaboration in a software engineering course	Marie Salomon, Kyle D. Chin, Reid Holmes et al.	2025	Proceedings of the 2025 ACM SIGPLAN International Symposium on SPLASH-E
Analyzing the communication patterns of different teammate types in a software engineering course project	Yanye Luther, Lindsey Nielsen, Logan Cadman et al.	2025	Proceedings of the 33rd ACM International Conference on the Foundations of Software Engineering
Assessing teamwork dynamics in software development projects	Santiago Berrezueta-Guzman, Ivan Parmacchi, Mohammad Kasra Habib et al.	2025	2025 IEEE Global Engineering Education Conference (EDUCON)
Ball point game: playing or learning agile project management?	Maria Lydia Fioravanti, Gustavo M. N. Avellar, Bruna Oliveira Romeiro et al.	2025	IEEE Access
Blended realities - higher education student reflections on acquiring skills for game creation in a project-based- and blended learning environment	Adam Palmquist, Joana Alves, Rene Baranyi et al.	2025	Games: Research and Practice
Chronicling consistency and parity of contributions to software engineering team projects	Kevin Buffardi, Rahul Bijoor, Aviral Kumar Srivastava et al.	2025	Proceedings of the 30th ACM Conference on Innovation and Technology in Computer Science Education V. 1
Community-oriented project-based learning for a web technologies course	Zheng Song, Sai Vikram Patnaik, Hayat Hachem et al.	2025	2025 IEEE Frontiers in Education Conference (FIE)

Continued on next page

Table 1 – continued from previous page

Title	Authors	Year	Venue
CTF gamified learning acceptance among Malaysian students	Nor Azlina Abd Rahman, Zaireeda Mohd Fauzee, Noris Ismail et al.	2025	2025 1st International Conference on Emerging Innovation and Digital Technology (ICEIDT)
Decoding student rating behaviors: a comparative study of frequent self and peer assessments in team projects	Yu Lu, André Van Der Hoek	2025	Proceedings of the 33rd ACM International Conference on the Foundations of Software Engineering
Developing leadership skills in software engineering students - a case study of a leadership development framework	Nicolas Nascimento, Afonso Sales, Rafael Chanin	2025	2025 IEEE Frontiers in Education Conference (FIE)
Embracing experiential learning: hackathons as an educational strategy for shaping soft skills in software engineering	Allysson Allex Araújo, Marcos Kalinowski, Maria Teresa Baldassarre	2025	2025 IEEE/ACM 37th International Conference on Software Engineering Education and Training (CSEE&T)
Emotion-centric AI interventions for experiential learning in higher education	Ayush Bharadwaj, Anshuman Sahu, L. Amit Kumar Das et al.	2025	2025 International Conference on Innovations in Intelligent Systems: Advancements in Computing, Communication, and Cybersecurity (ISAC3)
Empowering GenAI with SMCKI collaborative script in CSCL to enhance undergraduate students' collaborative problem-solving ability and learning performance	Ziwen Wei, Yi Zhang, Shuangting Ding et al.	2025	2025 International Conference on Educational Technology Management (ICETM)
Enhancing HCI capstone courses through diary study methods: feedback and insights from graduate-undergraduate collaboration	Jixiang Fan, Wei-Lu Wang, Natalie Andrus et al.	2025	Proceedings of the 7th Annual Symposium on HCI Education
Enhancing teamwork in project-based learning: challenges, reflections, and strategies for improvement	Aparajita Jaiswal, Paul Thomas, Prateek Jaiswal et al.	2025	2025 IEEE Frontiers in Education Conference (FIE)
Evaluating assessment practices in team-based computing capstone projects	Sara Hooshangi, Asma Shakil, Steve Riddle et al.	2025	Proceedings of the 2025 Working Group Reports on Innovation and Technology in Computer Science Education
Evaluating teamwork components in large undergraduate software engineering teams	Luka Dorić, Nikola Luburić, Jelena Slivka et al.	2025	ACM Transactions on Computing Education
Experimenting team-based learning in a large computer networks class	Paolo Casari, Sabrina Maniero, Andrea Rosani et al.	2025	IEEE Transactions on Education
Exploring team dynamics in software development: personality traits and collaborative programming	Samantha Jiménez, Camilo Caraveo, Leticia Cervantes Huerta et al.	2025	2025 13th International Conference in Software Engineering Research and Innovation (CONISOFT)
Exploring topic selection in undergraduate capstone courses: impacts on engagement, teamwork, and learning outcomes	Alex Modarresi, Vaishnavi Sen	2025	Proceedings of the 26th ACM Annual Conference on Cybersecurity & Information Technology Education
Game studio simulator: integrating industry practice and formative individual assessment in collaborative game development capstone projects	Charlotte Pierce, Clinton J. Woodward, Andrew Trevillian et al.	2025	ACM Transactions on Computing Education

Continued on next page

Table 1 – continued from previous page

Title	Authors	Year	Venue
Group assignments and support aimed to develop student teamwork skills and a positive attitude towards teamwork in computer science higher education	Cristina Adriana Alexandru	2025	Proceedings of the 9th Conference on Computing Education Practice
Hands-on learning in database programming education: exploring student experiences through thematic analysis of project reflections	Nurhafizah Mahri, Nur Leyni Nilam Putri Junurham	2025	2025 10th International Conference on Information and Communication Technology for the Muslim World (ICT4M)
How soft skills shape first-year success in higher education	Kerstin Andree, Santiago Berrezueta-Guzman, Stephan Krusche et al.	2025	2025 IEEE 4th German Education Conference (GECon)
Human ants are beneficial for team performance	Iflaah Salman, Alvaro Francisco Gil, Juan Garbajosa et al.	2025	2025 32nd Asia-Pacific Software Engineering Conference (APSEC)
Improving software engineering team communication through stronger social networks	April Clarke, Tanja Mitrović, Fabian Gilson	2025	2025 IEEE/ACM 37th International Conference on Software Engineering Education and Training (CSEE&T)
Influence of individual image on team dynamics in IT-related undergraduate projects: investigating the hue effect	B.M.G.S.T.S.K. Wickramasinghe, K.D.B. Aratthanage, W.A.L.P. Fernando et al.	2025	2025 9th SLAAI International Conference on Artificial Intelligence (SLAAI-ICAI)
Integrating soft skills training into your course through a collaborative activity	Géraldine Brieven, Marcia Moraes, Dieter Pawelczak et al.	2025	Proceedings of the 56th ACM Technical Symposium on Computer Science Education V. 1
Investigating the role of the instructional team in enhancing student teamwork experiences in an introductory cybersecurity course	Aparajita Jaiswal, Paul Thomas, Sudip Vhaduri et al.	2025	2025 IEEE Frontiers in Education Conference (FIE)
Learning to code together: a look into student projects in an object-oriented programming course	Marina Lepp, Eva Luude	2025	2025 IEEE International Conference on Teaching, Assessment, and Learning for Engineering (TALE)
LLM contribution summarization in software projects	Rafael Corsi Ferrão, Fabio De Miranda, Diego Pavan Soler et al.	2025	2025 IEEE Frontiers in Education Conference (FIE)
Navigating multi-semester service learning in software engineering: strategies for success	Chad A. Williams, Stan Kurkovsky, Mikey Goldweber et al.	2025	2025 IEEE Frontiers in Education Conference (FIE)
Peer learning through distributed computing research for diverse undergraduate students	Pradeeban Kathiravelu, David Moxley	2025	2025 IEEE International Performance, Computing, and Communications Conference (IPCCC)
Project-based learning for teaching natural language processing in undergraduate computing	Thiago Goulart De Oliveira, Henrique Moallem Paiva, Victor Takashi Hayashi	2025	2025 International Symposium on Accreditation of Engineering and Computing Education (ICACIT)
Project-based robotics 4.0 education using microcontrollers to enhance technical and programming competencies	József Udvaros, Veronika Gabal'ová	2025	2025 IEEE 25th International Symposium on Computational Intelligence and Informatics (CINTI)
Quality of collaborative learning discovered by data mining and visualization	Yixuan Wang, Nanlin Jin, Ho-Pun Lam	2025	2025 5th International Conference on Artificial Intelligence and Education (ICAIE)

Continued on next page

Table 1 – continued from previous page

Title	Authors	Year	Venue
Regression testing skill transfer to industry: a preliminary study in higher education	Andrada-Mihaela-Nicoleta Moldovan, Andreea Vescan	2025	2025 40th IEEE/ACM International Conference on Automated Software Engineering Workshops (ASEW)
Remedying low student performance in introductory programming: a case study on role assignment in academic teams	Nikolaos Spanoudakis, Evangelia Krasadaki	2025	2025 6th International Conference in Electronic Engineering & Information Technology (EEITE)
Scaffolded projects for the social good: broadening participation in service learning for computing curricula	Stan Kurkovsky, Michael Goldweber, Chad A. Williams et al.	2025	Proceedings of the ACM Global Computing Education Conference 2025 - Volume 1
Scaffolding creative thinking in project-based learning: a practical use of scrum and generative AI	Anna Lyza Felipe, James Harland, Duy Dang et al.	2025	2025 IEEE International Conference on Teaching, Assessment, and Learning for Engineering (TALE)
Scaffolding online capstone courses for improving data science education via comparative analysis of online and In-person capstone courses	Xinjie Lan	2025	2025 IEEE Digital Education and MOOCS Conference (DEMOcon)
Self-assessment and the development of socio-emotional competencies in first-year computing students	Julia Stateri, Filipe G. De Souza N. Da Silva, Rajesh Montoya Rani	2025	2025 International Symposium on Accreditation of Engineering and Computing Education (ICACIT)
Software engineering courses through the lens of an education intelligence tool	Manuel Stöger, Johann Grabner, Mario Bernhart et al.	2025	2025 IEEE/ACM 37th International Conference on Software Engineering Education and Training (CSEE&T)
Software engineering team project courses with industrial customers: students' insights on challenges and lessons learned	Nayla Nasir, Muhammad Usman, Jürgen Börstler et al.	2025	Journal of Systems and Software
Stimulating brainstorming activities with generative AI in higher education	Jérémy La Scala, Sonia Sahli, Denis Gillet	2025	2025 IEEE Global Engineering Education Conference (EDUCON)
Supporting brainstorming activities with bots in software engineering education	Juan Carlos Farah, Jérémy La Scala, Sandy Ingram et al.	2025	2025 IEEE/ACM International Workshop on Bots in Software Engineering (BotSE)
Teaching of transferable teamwork competencies in higher education: development of the TWC training protocol©	Cynthia G. Campbell, Iryna Babik, Stephen J. Crowley	2025	Frontiers in Education
Team composition in software engineering capstone courses: the role of pre-existing relationships	Idunn Stabell, Camilla Kielland Lund, Viktoria Stray et al.	2025	Proceedings of the 2025 29th International Conference on Evaluation and Assessment in Software Engineering Companion
Team versus individual projects in introductory programming: effects on performance, variability, and retention	Hyunchul Joh	2025	IEEE Access
Teamwork in agile software development: a mixed-method study of gender diversity and collaboration dynamics	Viktoria Stray, Idunn Stabell, Gyda Elisa Sæter et al.	2025	Information and Software Technology
Teamwork quality in agile capstone courses: insights from 790 software engineering students	Camilla Kielland Lund, Idunn Stabell, Viktoria Stray et al.	2025	Proceedings of the 33rd ACM International Conference on the Foundations of Software Engineering

Continued on next page

Table 1 – continued from previous page

Title	Authors	Year	Venue
The collaborative disposition: what students say	Renée McCauley, Mihaela Sabin, Bonnie K. MacKellar et al.	2025	2025 IEEE Frontiers in Education Conference (FIE)
The development of reflective practice on a work-based software engineering program: a longitudinal study	Matthew Barr, Syed Waqar Nabi, Oana Andrei	2025	2025 IEEE/ACM 37th International Conference on Software Engineering Education and Training (CSEE&T)
The influence of software engineering education on developing teamwork skills in both genders	María Cecilia Bastarrica, Jocelyn Simmonds	2025	IEEE Access
The role of project-based learning with activity theory in teaching effectiveness: evidence from the internet of things course	Zhicheng Dai, Yue Yang, Zengzhao Chen et al.	2025	Education and Information Technologies
Towards emotionally intelligent software engineers: understanding students' self-perceptions after a cooperative learning experience	Allysson Alex Araújo, Marcos Kalinowski, Matheus Paixao et al.	2025	2025 IEEE/ACM 18th International Conference on Cooperative and Human Aspects of Software Engineering (CHASE)
Two-stage CS exams: a multi-institutional study of course and exam feature impacts on student perceptions and performance	Celine Latulipe, John Anvik, Kevin Lin et al.	2025	Proceedings of the 2025 Working Group Reports on Innovation and Technology in Computer Science Education
Understanding students' mindset while reflecting on their teamwork experiences	Shaimaa Ali, Minha R. Ha	2025	2025 World Engineering Education Forum - Global Engineering Deans Council (WEEF-GEDC)
Understanding students' perceptions of human dispositions in computing education	Teresa Pereira, John Impagliazzo	2025	2025 IEEE Frontiers in Education Conference (FIE)
Understanding the effect of agile practice quality on software product quality	Sherlock A. Licorish	2025	IEEE Transactions on Software Engineering
Use of search tools in software development: a study of microservice-based team projects:	Yi Lau, Christian Koh, Lingxiao Jiang	2025	Proceedings of the 17th International Conference on Computer Supported Education
Using scrum to improve student teamwork in a project-based hybrid learning setting	Essa Imhmed, Edgar Ceh-Varela, Mustafa Elfturi	2025	2025 IEEE Frontiers in Education Conference (FIE)
Variation in engagement behaviors among student-centered pedagogies	Patricia B. Campbell, Clif Kussmaul, Chris Mayfield et al.	2025	Proceedings of the 56th ACM Technical Symposium on Computer Science Education V. 1
Visualizing frequent peer assessment results to uncover problems in team dynamics	Yu Lu, André Van Der Hoek	2025	2025 IEEE/ACM 37th International Conference on Software Engineering Education and Training (CSEE&T)
Visualizing team progress and contributions in student projects with teamcast	Anmol Vilas Deshpande, André Van Der Hoek	2025	2025 IEEE/ACM 37th International Conference on Software Engineering Education and Training (CSEE&T)
Web platform with scrum approach to improve academic performance in a private higher education institution in Peru	Jean Franco Vega-Rodriguez, Diego Torres-De-La-Cruz, Daniel Burga et al.	2025	2025 IEEE XXXII International Conference on Electronics, Electrical Engineering and Computing (INTERCON)

Continued on next page

Table 1 – continued from previous page

Title	Authors	Year	Venue
Who introduces and who fixes? Code quality dynamics in collaborative embedded systems projects	Rafael Corsi Ferrão, Igor Dos Santos Montagner, Roldofo Azevedo	2025	2025 IEEE Frontiers in Education Conference (FIE)
WIP: enhancing computer programming courses with hardware and simulation projects	Asad Azemi	2025	2025 IEEE Frontiers in Education Conference (FIE)
WIP: fostering effective teamwork in undergraduate research based course	Priyanka Kumar, Panhapiseth Lim, Saniya Movahed et al.	2025	2025 IEEE Frontiers in Education Conference (FIE)
WIP: implementing scrum methodology in engineering capstone design courses	Haifeng Wang	2025	2025 IEEE Frontiers in Education Conference (FIE)
Collaboration under competition empirical impacts on motivation and teamwork in a multiplayer programming game	Ratchanon Nobnop, Jetmanus Ratisaksoontorn, Pakakrong Panthanan et al.	2026	2026 Joint International Conference on Digital Arts, Media and Technology with ECTI Northern Section Conference on Electrical, Electronics, Computer and Telecommunication Engineering (ECTI DAMT & NCON)
Competing or collaborating? The role of hackathon formats in shaping team dynamics and project choices	Sadia Nasrin Tisha, Md Nazmus Sakib, Sanorita Dey	2026	Proceedings of the 57th ACM Technical Symposium on Computer Science Education V.1
Establishing the criteria for an agile inclusive learning environment (AILE) in higher education	Itzel Berenice Guerrero Alfaro, Jaime Muñoz-Arteaga, Gabriela Citlalli López Torres et al.	2026	IEEE Revista Iberoamericana de Tecnologías del Aprendizaje
Improving professional dispositions in computing curriculum using sequential peer assessment	Dhaval K. Patel, Ayush N. Patel, Raj N. Dave	2026	Proceedings of the 57th ACM Technical Symposium on Computer Science Education V.1
Integrating human skills and debates into computing education: an experience report	Eric Fouh	2026	Proceedings of the 28th Australasian Computing Education Conference
Integration of learning framework for software engineering education	Mira Balaban, Shahaf S. Shperberg, Arnon Sturm	2026	ACM Transactions on Computing Education
LLMs integration in software engineering team projects: roles, impact, and a pedagogical design space for AI tools in computing education	Ahmed Kharrufa, Sami Alghamdi, Abeer Aziz et al.	2026	ACM Transactions on Computing Education
Methodology for monitoring internal team climate in agile educational projects using weighted evaluation systems	Héctor Cornide-Reyes, Cristián Durán, Juan F. Olivares-Pacheco et al.	2026	IEEE Access
Promoting successful projects in a software engineering capstone course	María Cecilia Bastarrica, Cristian Salazar	2026	Anais do XXIX Congresso Ibero-Americano em Engenharia de Software (CIBSE 2026)
Results of a gamification university experience for teaching web programming: the “alice in codeland” project	Daniele Zolezzi, Luca Martini, Saverio Iacono et al.	2026	IEEE Transactions on Education
Self-selected groups vs. Random groups: an analysis of student engagement, achievement, and preferences in collaborative learning	Ati Suci Dian Martha, Sri Widowati, Arinza Aurelvia et al.	2026	Collaboration technologies and social computing
Soft skills and group performance in higher education: a quantitative study of student teams in software engineering	Kelian Schekkerman, Andrea Capiluppi	2026	Computer Applications in Engineering Education

Continued on next page

Table 1 – continued from previous page

Title	Authors	Year	Venue
Teaching AI competencies: experiences from coaching interdisciplinary teams to develop AI prototypes	Aidin Azamnouri, Dominik Hörauf, Brigitte Schönberger et al.	2026	Computers and Education: Artificial Intelligence
Teamwork in software engineering education: exploring team dynamics and challenges across course projects	Yu Lu, Niklas Meißner, Matthew Bietz et al.	2026	Proceedings of the IEEE/ACM 48th International Conference on Software Engineering
The boundary-spanning assistant: understanding the role and usage patterns of LLMs in project-based software engineering	Anh Nguyen-Duc, Kai-Kristian Kemell, Aparna Chirumamilla	2026	Proceedings of the IEEE/ACM 48th International Conference on Software Engineering
Using peer evaluations and team contracts in software engineering team project courses	Nayla Nasir, Muhammad Usman, Usman Nasir	2026	Proceedings of the IEEE/ACM 48th International Conference on Software Engineering
When emotions matter: speech-based emotion detection in capstone project retrospective meetings while working hybrid	Wardah Naeem Awan, Maria Paasivaara, Peter A. Gloor	2026	Proceedings of the IEEE/ACM 48th International Conference on Software Engineering
Who plays which role when? Communication role dynamics for peer recognition and team performance prediction	Yifan Song, Wenxuan Wendy Shi, Brian Bailey et al.	2026	Proceedings of the 64th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)

Continued on next page